

LEARNER GUIDE

Unit Standard Title: CARRY OUT BASIC FIRST AID TREATMENT IN THE
WORKPLACE

Unit Standard No: 116534

Unit Standard Credits: 2

NQF Level: 3

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This material has been compiled based on the **UNIT STANDARD 116534: Carry out basic first aid treatment in the workplace**

The program has been accredited by SASSETA in the Republic of South Africa.

Material compiled by:

Training Dynamics Africa

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1. Introduction

This learning program is part of a complete qualification. For more details concerning the complete qualification please contact your service provider or alternatively the office of the SASSETA

2. Purpose of this learning program

This unit standard is for persons who are responsible to apply first aid in a working place. Persons credited with this standard are able to:

- ♦ Describe first aid equipment and explain basic application
- ♦ Assess the accident scene, report and make safe
- ♦ Prioritize the casualties and treatment

2.1 Target group

This program is compiled for the following **target group**:

- Security members
- South African Defense Force members
- South African Police Force members
- Correctional Services
- Individuals who wishes to complete the NQF level 3 National Certificate in security practices.

3. Standards and qualifications

It is important that you as the learner understand and realize that this training program is not presented in isolation, but that it was developed based upon nationally recognized standards known as unit standards.

Unit standards are the “building blocks” of qualifications. All qualifications are plotted on the National Qualifications Framework (NQF).

REMEMBER.....

Unit standards comprises of outcomes. An outcome is a statement that describes the required competency that must be demonstrated by the learner on successful completion of a training intervention.

4. Assessments

In order to assess whether a learner can actually demonstrate the desired outcomes, assessment criteria are included in the unit standard. Each outcome has its own set of assessment criteria.

**REMEMBER.....**

The assessment criteria describe the evidence that is needed that will show that the learner has demonstrated the outcome correctly.

Kindly refer to the unit standard attached hereto for the assessment criteria.

It is of utmost importance that the learner fully understands the assessment criteria as listed in the unit standard, as it is the only way in which the learner will know what he will be assessed against.

The **final or summative assessment** is the most important aspect of this training program. It is during this process that the learner will be declared competent or not yet competent.

The learner will know exactly how he will be assessed, and when and where he will be assessed. All of these details must be obtained from the training provider where the learner enrolled for his program.

5. Range statements

**REMEMBER.....**

Also included in the unit standard are the range statements in support of the assessment criteria.

The range statements indicate detailed requirements of the assessment criteria.

6. The learning guide

The learner guide is included in this material under various learning units. The learner guide has been designed in such a manner that the learner is guided in a logical way through the learning material and requirements of the unit standard.

On completion of this skill program, the learner will be assessed against the assessment criteria as stipulated in the unit standard.

Each of the modules consists of learner outcomes to be achieved during training and development interventions, work related experience or at your own time. The learner outcomes will enable you to achieve the required level of confidence and competence in order to undergo the summative (final evaluations) assessment.

The learner needs to assess his own knowledge and skill throughout the training process by completing the learner workbook. With the completion of the workbook you start collecting evidence to proof your competence. The learner workbook will be assessed (FORMATIVE ASSESSMENT)

TAKE NOTE

The learner guide will remain the property of the learner once the LEARNING PROGRAM has been completed.

TAKE NOTE

- The best results will be obtained if you start with Study Unit 1 in the Learner Guide, and work your way through it, study unit by study unit.
- All learning outcomes are vital and must be studied (and exercised for all practical requirements) thoroughly to ensure that enough evidence for all specific outcomes is generated.

7. Learning assumed to be in place

It is assumed that a learner which registers for this program have completed a program in communication at NQF Level 3 or equivalent.

It is imperative to understand that this program are only one of the building blocks to complete Skills program 1 The complete list of unit standards for Skills program appears on the next page.

7.1 Security program matrix

SKILLS PROGRAM 1: SASSETA E			
244177	Conduct a security patrol in area of responsibility	Level 3	7 Credits
244184	Apply legal aspects in a security environment	Level 3	8 Credits
244176	Use security equipment	Level 2	2 Credits
244179	Handle complaints and problems	Level 3	6 Credits
244181	Perform hand over and take over responsibilities	Level 3	2 Credits
244182	Give evidence in court	Level 3	4 Credits
12484	Perform basic fire fighting	Level 2	4 Credits
116534	Carry out basic first aid treatment in the workplace	Level 3	2 Credits

Patrol Security officer/Grade E

SKILLS PROGRAM 2: SASSETA D			
244189	Conduct access and egress control	Level 4	7 Credits
242825	Conduct evacuations and emergency drills	Level 4	4 Credits
11505	Identify, handle and defuse security related conflict	Level 4	12 Credits
117705	Demonstrate knowledge of the Firearms Control Act 2000 (Act No. 60 of 2000)	Level 3	3 Credits
113924	Apply basic business ethics in a work environment	Level 2	2 Credits
119465	Write/present/sign texts for a range of communicative contexts	Level 3	5 Credits
114979	Operate a computer workstation in a business environment	Level 3	2 Credits

Access control officer/Grade D

SKILLS PROGRAM 3: SASSETA C			
113909	Coach a team member in order to enhance individual performance in work environment	Level 3	5 Credits
13912	Apply knowledge of self and team in order to develop a plan to enhance team performance	Level 3	5 Credits
244578	Describe how to manage reactions arising from a traumatic event	Level 3	2 Credits
113852	Apply occupational health, safety and environmental principles	Level 3	10 Credits
13936	Outline the legal environment of a selected industry	Level 3	2 Credits
119472	Accommodate audience and context needs in oral/signed communication	Level 3	5 Credits
11983	Compile and administer a case docket for investigation purposes	Level 5	6 Credits

Asset & reaction officer/Grade C

116534: Carry out basic first aid treatment in the workplace

8. Preparation of summative assessment

After you have worked through the **LEARNER GUIDE** and you are satisfied that all theoretical and practical requirements/ evidence, as stipulated by the UNIT STANDARD, can be met, you need to make your own necessary arrangements with the assessor and/ or facilitator. The training establishment will then arrange a suitable date for you to attend your pre-assessment meeting and then arrange for a suitable date to complete the final assessment.

8.1 RPL assessment



The assessment of RPL learners will be conducted in the same way as for those of new learners. The assessment pack is exactly the same and will therefore be used for new learners as well as RPL Learners. It must however be noted that learners who are applying for RPL must provide proof of previous learning and subject related experience prior to the assessment.

This proof or evidence can be in the format of certified copies (certificates) of previous learning programs that have been attended.

All the evidence will be assessed and authenticated before a learner will be allowed to enroll for an RPL program.

9. Roles and responsibilities

9.1 Facilitator/instructor

- Prepare to facilitate.
- Transfer of knowledge and skills

9.2 The role of the assessor

- Plan for the assessment
- Prepare the learner for the assessment

9.3 The role of the Learner

- Prepare for assessment.
- Accept the responsibility of learning.
- Accept the responsibility to produce/deliver evidence.

STUDY UNIT 1

LEGAL COMPLIANCE

LEARNING OUTCOMES

On completion of this study unit the student will....



- ♦ Define the term first aid.
- ♦ Demonstrate knowledge of the elements of first aid.
- ♦ Explain the first aid process.
- ♦ List and explain the responsibilities of a first aider.
- ♦ Demonstrate knowledge of the ethics of first aid by focusing on:
 - Medical ethics
 - Social norms and standard
 - Knowledge and skill
- ♦ Demonstrate knowledge of the importance of regular monitoring and recording of vital functions.
- ♦ Demonstrate knowledge and skills of the procedure to be followed for handing a patient over to advance care.
- ♦ Demonstrate skills regarding the procedure to be followed when vital functions are monitored and recorded.

1. Principles and ethics of first aid

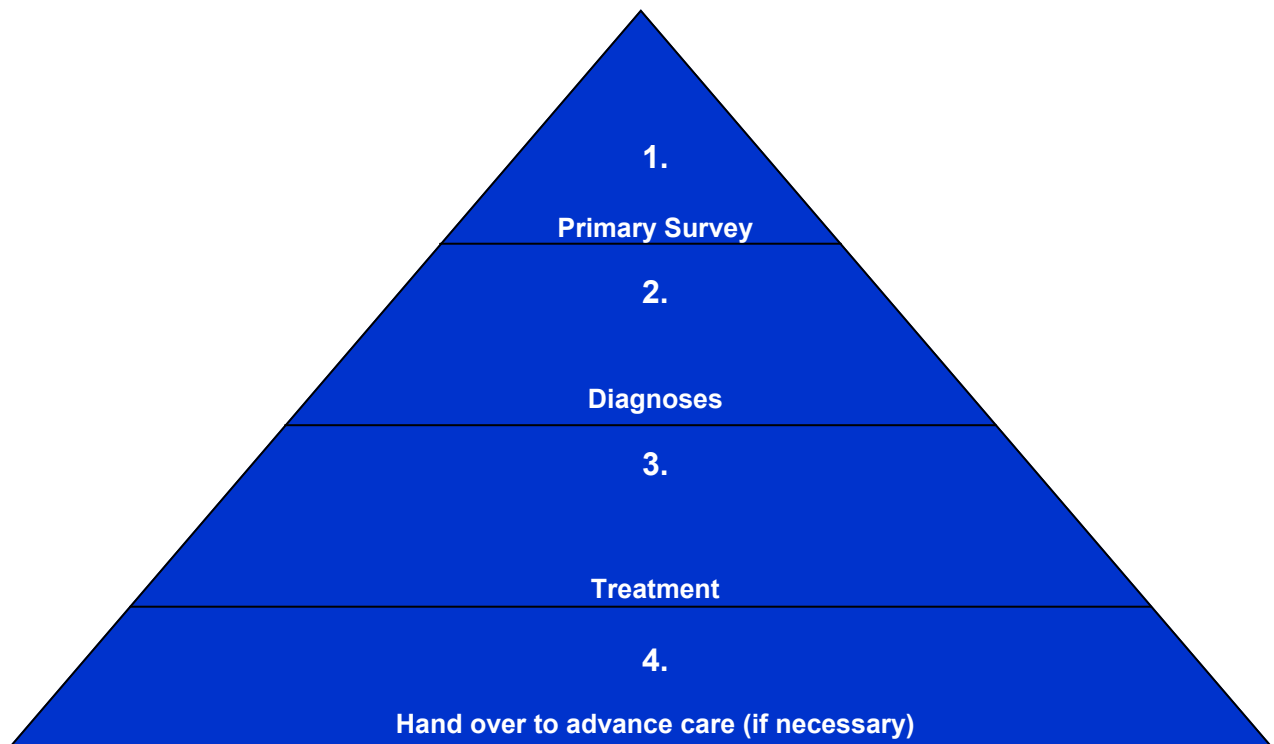


First aid is the effective application of approved measures and the utilizing of available resources to assist a person who has suffered trauma or acute illness.

1.1 Elements of First Aid

- ♦ To save lives
- ♦ To relieve pain and suffering and prevent further injuries and suffering
- ♦ To promote healing

2. The First Aid Process



3. Pre-Hospital Emergency Care

DESCRIPTION	LEVEL
GROUP 1	
Basic Life Support (BLS)	Level 1, 2, 3 & 4
GROUP 2	
Intermediate Life Support (ILS)	Level 5 & 6
GROUP 3	
Advance/Critical Life Support (ALS)	Level 5 & 6



First Aid Certificates expire within 3 years

4. The responsibility of the First-aider

4.1 Towards the patient

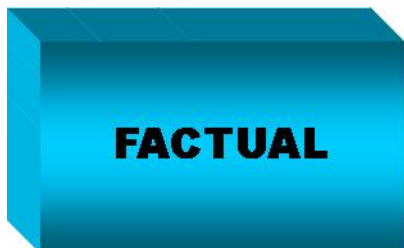
- ♦ The First-aider must act within his/her own levels of skill and training.
- ♦ The First-aider must respect his/her patients as well as the patient's privacy.
- ♦ First-aider may not take part in any harmful practices.

4.2 Towards the Public

- ♦ The First-aider has a responsibility towards the safety of any bystanders at a scene.
- ♦ The First-aider has a responsibility towards public awareness concerning health, safety and First Aid principles.

5. The ethics of First Aid

5.1 Medical Ethics



Adopted from the Hippocratic Oath

TAKE NOTE

- ♦ To respect the personal secrets of my patient.
- ♦ To respect my colleagues. (medical profession)
- ♦ Not to permit religion, nationality, race, party politics or social standing to influence my conduct and actions.

5.2 Social Norms and Standards

- ♦ Respect the social norms and standards of your patients, their families and bystanders.

5.3 Knowledge and Skills

- ♦ The First-aider needs keep him/herself up to date their knowledge.
- ♦ The First-aider needs read and attend refresher training, workshops and seminars on a regular basis.
- ♦ The First-aider needs practice regularly.
- ♦ Debriefing must be done on a regular basis.

6. Regular monitoring and recording of vital functions**CRITICAL**

**ALWAYS MONITOR YOUR PATIENT REGULARLY
AND KEEP RECORD OF HIS PROGRESS**

6.1. Important Factors

- ♦ State of consciousness (awake, confused, semi-conscious or unconscious)
- ♦ Breathing
- ♦ Pulse
- ♦ Skin temperature

7. Procedure for handing a patient over to advance care

You must provide the following details:

- ♦ Your own details. (name and first aid qualifications)
- ♦ History and details of incident.
- ♦ Signs and symptoms.
- ♦ Your own assessments and diagnoses.
- ♦ Treatment given.
- ♦ Patient's current vital signs. (see par. h above)
- ♦ Patient's progress. (improved or deteriorated)

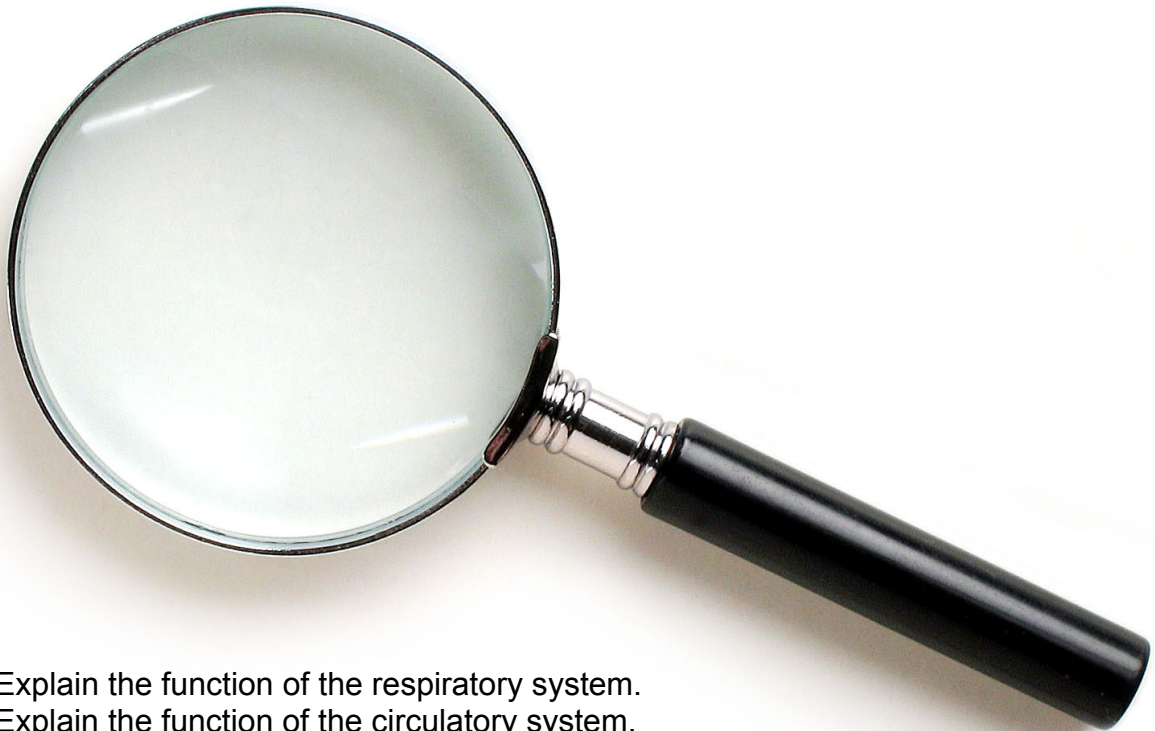
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STUDY UNIT 2

BASIC ANATOMY AND PHYSIOLOGY OF THE HUMAN BODY

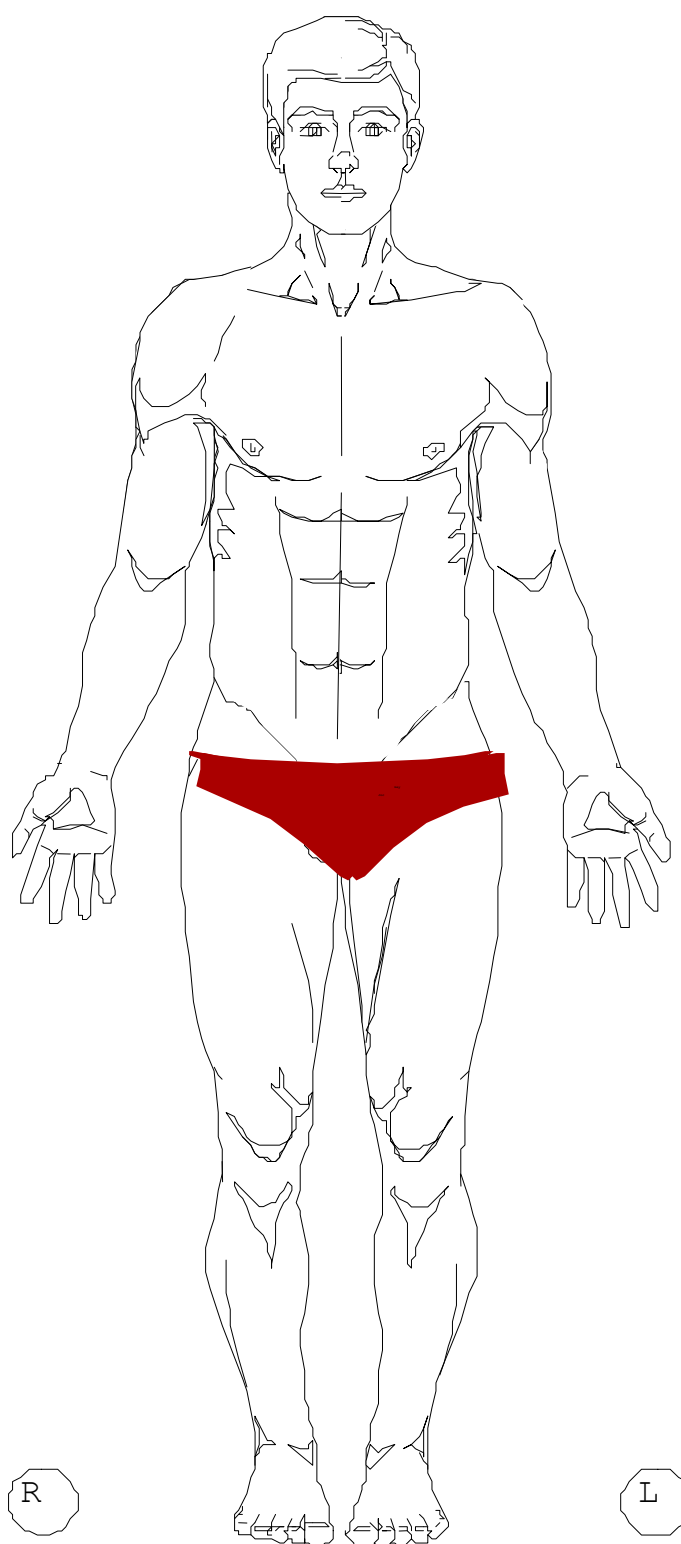
LEARNING OUTCOMES

On completion of this study unit the student will....



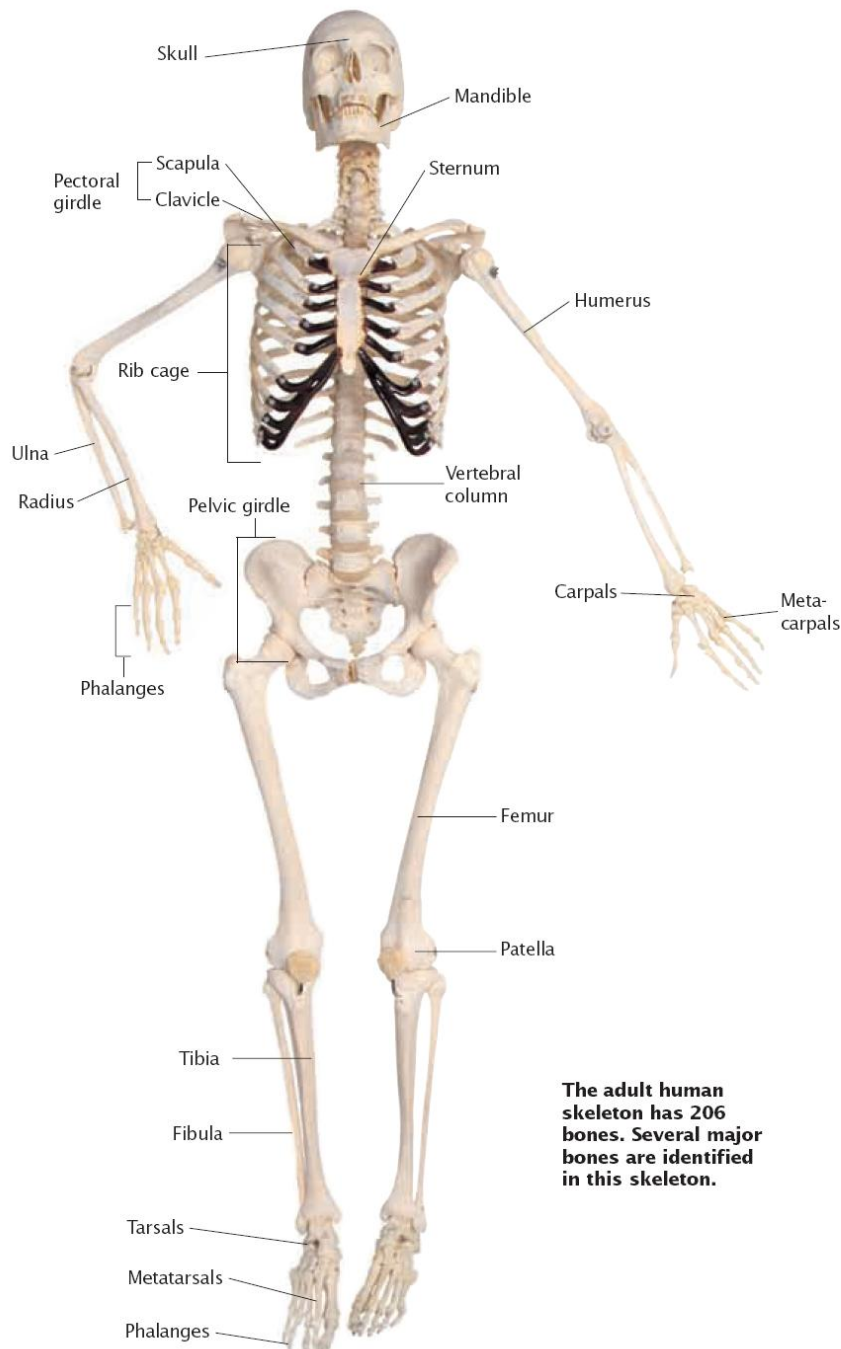
- ♦ Explain the function of the respiratory system.
- ♦ Explain the function of the circulatory system.
- ♦ Explain the function of the nervous system.
- ♦ Explain the function of the skeleton.
- ♦ Explain the function of muscles.
- ♦ Explain the function of the spinal cord.
- ♦ Demonstrate the key points where a pulse can be felt.
- ♦ Demonstrate the method of feeling a pulse.

1. The anatomical position



2. The Skeleton

Human Skeleton

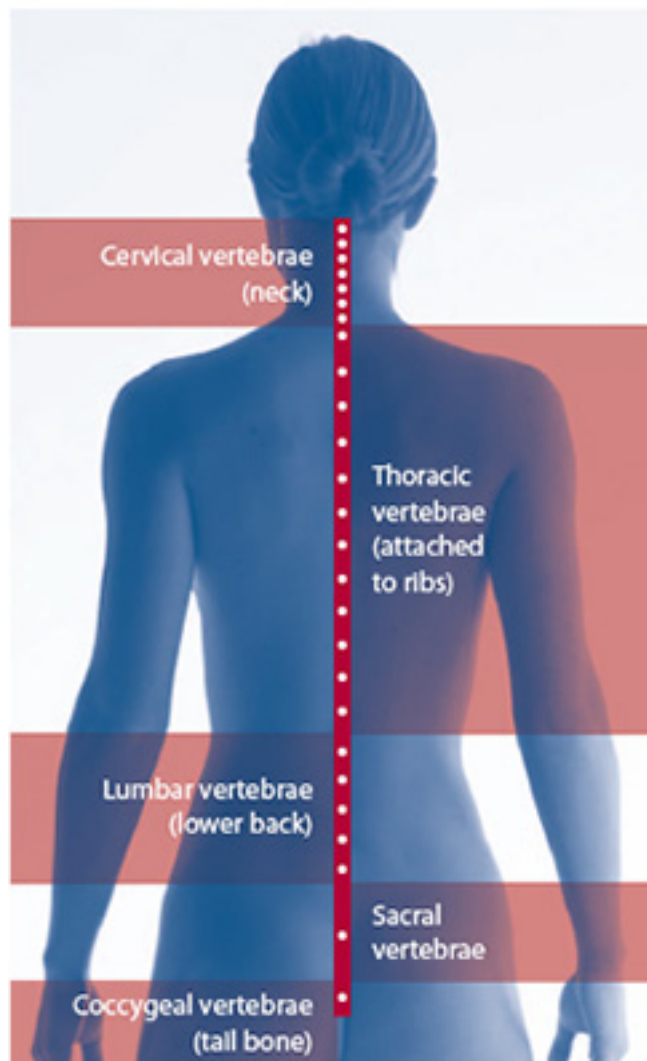


The adult human skeleton has 206 bones. Several major bones are identified in this skeleton.

2.1 Function of the skeleton

- ♦ Supports the whole body.
- ♦ Protects organs. (Brain, heart etc.)
- ♦ Movement (muscles attach to skeletal bones).

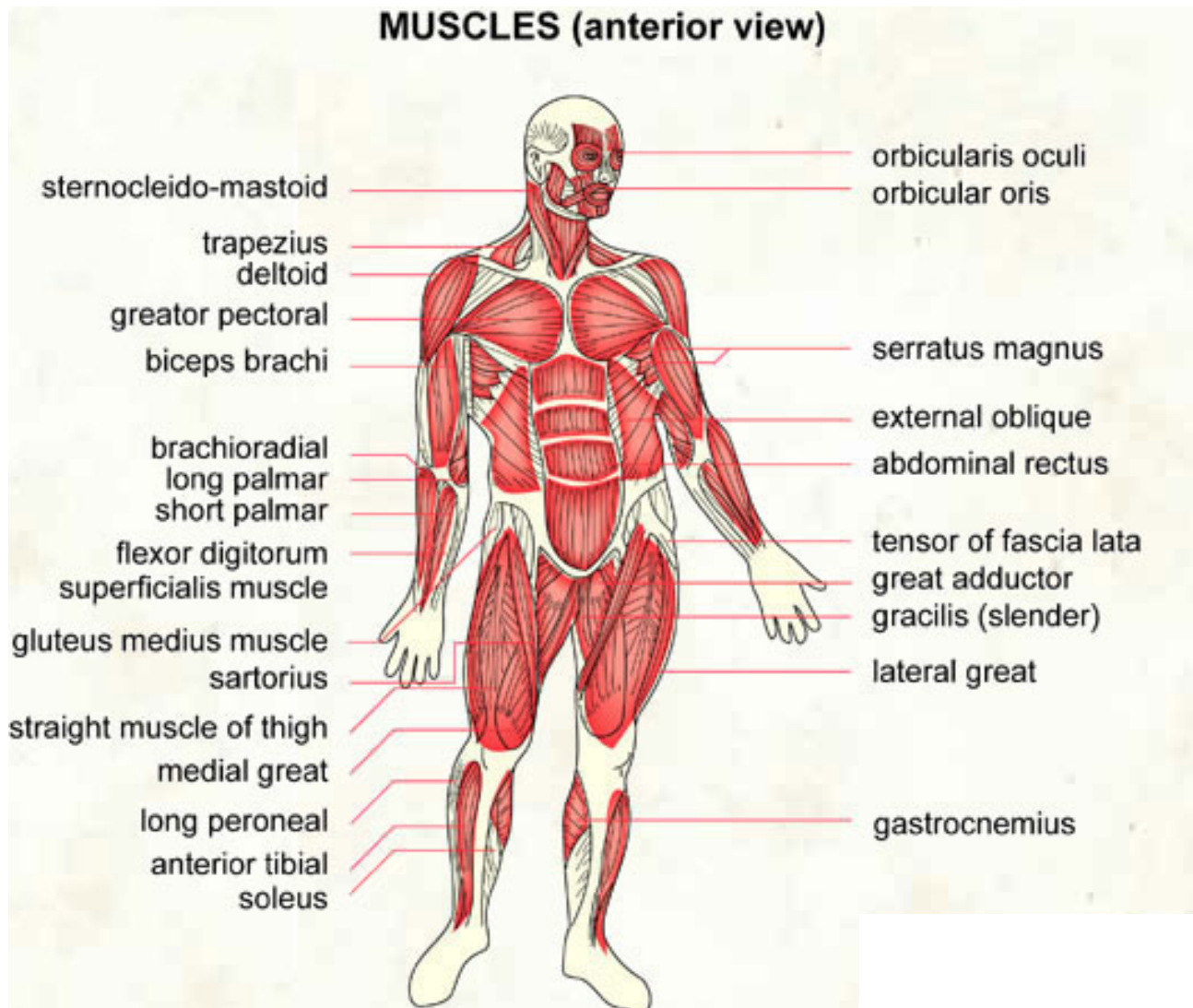
3. The Spinal column



3.1 Functions of the spinal column

- ♦ Protects the spinal cord
- ♦ Movement
- ♦ Supports torso and skull
- ♦ Shock absorbing

4. Muscles



- ◆ Muscles (anterior view): fibrous organs that produce motion by contracting.
- ◆ Orbicularis oculi: circular muscle of the eyelids.
- ◆ Orbicular oris: circular muscle of the lips.
- ◆ Serratus magnus: finger-shaped muscles between the ribs.
- ◆ External Oblique: muscle whose fibres are oblique in relation to the body when it is standing.
- ◆ Abdominal rectus: abdominal muscle with vertical fibres.
- ◆ Tensor of fascia lata: muscular membrane that stretches and tightens.
- ◆ Great adductor: muscle whose movement runs parallel to the plane of symmetry of the body.
- ◆ Gracilis (slender): vertical muscle of the inner thigh.
- ◆ Lateral great: large muscle of the outer thigh.
- ◆ Gastrocnemius: the two muscles of the calf.
- ◆ Soleus: extensor muscle of the foot.
- ◆ Anterior tibial: muscle of the leg below the knee.
- ◆ Long peroneal: muscle below the peroneals and above the metatarsals.

- ♦ Medial great: large muscle of the outer thigh.
- ♦ Straight muscle of thigh: vertical muscle above the knee.
- ♦ Sartorius: muscle that pivots the lower leg on the thigh and the thigh on the pelvis.
- ♦ Gluteus medius muscle: muscle of the buttock.
- ♦ Flexor digitorum superficialis muscle: muscle that flexes the fingers.
- ♦ Short palmar: small muscle controlling the palm of the hand.
- ♦ Long palmar: large muscle controlling the palm of the hand.
- ♦ Brachioradial: muscle used to rotate the hand.
- ♦ Biceps brachi: an arm muscle with two points of attachment.
- ♦ Greater pectoral: bulky chest muscle.
- ♦ Deltoid: triangular shoulder muscle used for abduction of the arm.
- ♦ Trapezius: back muscle between the scapula and the spinal column.
- ♦ Sternocleido mastoid: neck muscle connecting the sternum to the clavicle and relative to the mastoid process.

4.1 Types

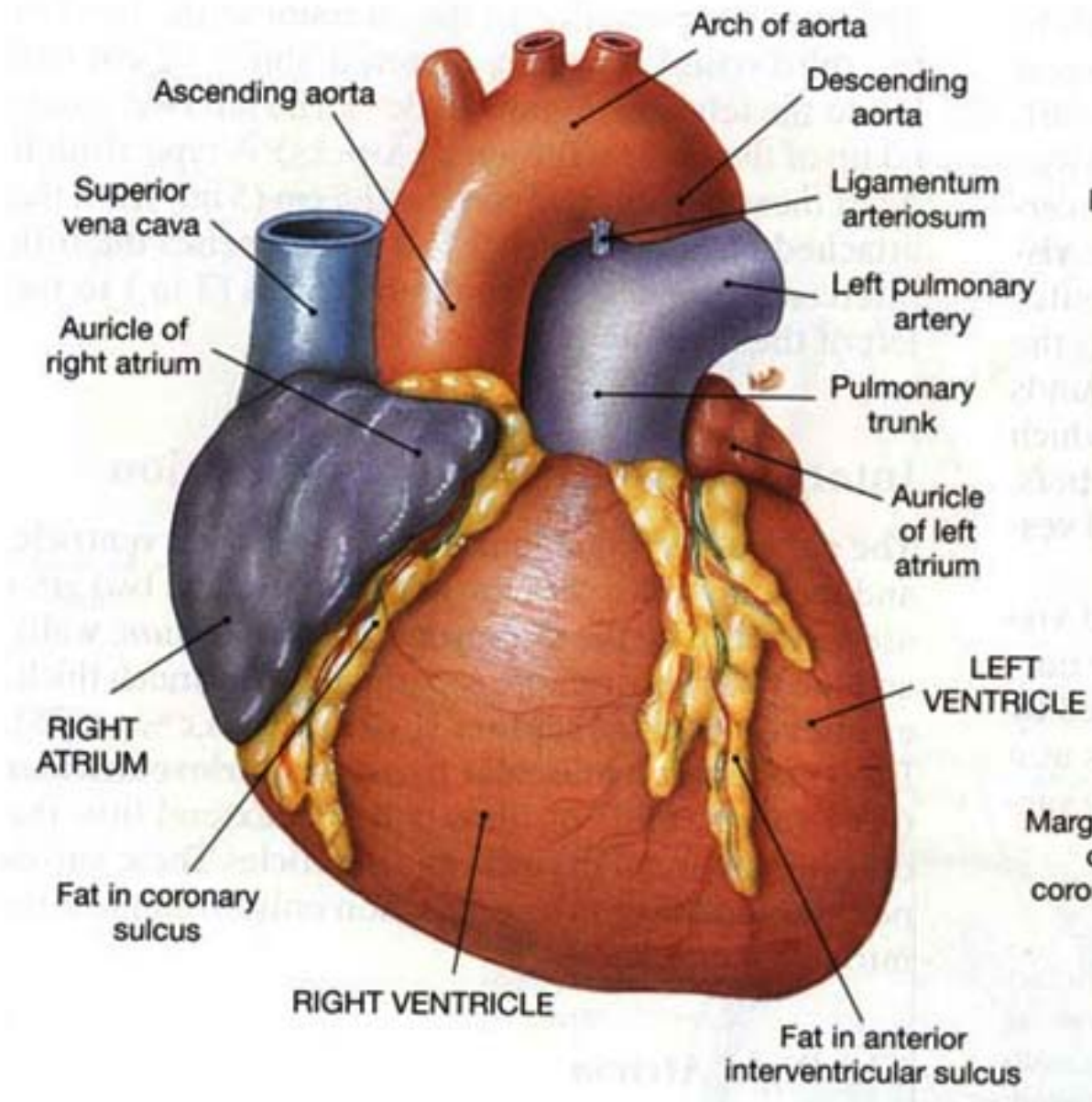
- ♦ Skeletal muscles: Attached to skeleton - voluntary
- ♦ Smooth muscles: Blood vessels and intestines - involuntary
- ♦ Heart muscle: Involuntary

4.2 Functions of muscles

- ♦ Movement
- ♦ Food processing
- ♦ Blood transport

5. The Circulation System

5.1 The Heart



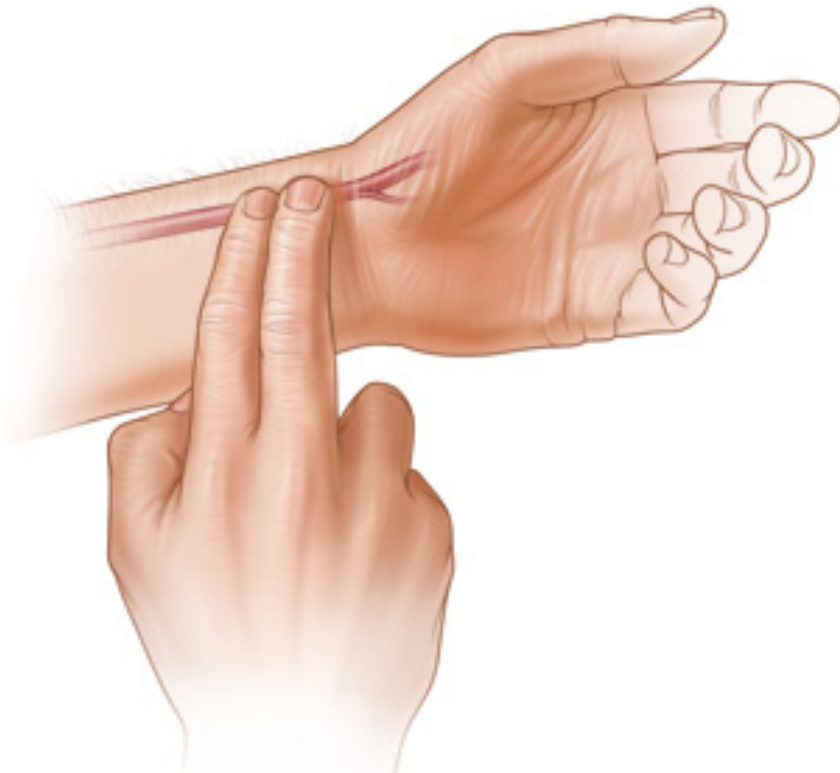
5.2 Arteries, veins and capillary vessels

- ♦ **Arteries:** Transport blood **away from** the heart.
- ♦ **Veins:** Transport blood **towards** the heart.
- ♦ **Capillary blood vessels:** Link arteries and veins.

5.3 Blood volumes

- ♦ **An adult male:** 6 to 7 liters
- ♦ **An adult female:** 5 to 6 liters
- ♦ **An infant:** 350 ml.

6. The Pulse



- ♦ Carotid: Unconscious
- ♦ Radial: Conscious
- ♦ Brachial: Infants
- ♦ Capillary refill (Cannot be felt (To check circulation only))

6.1 Feeling the pulse



WHY

Pulse measurement is performed to determine your heart rate and rhythm.

Your pulse is caused by your heart beating. As your heart beats and forces blood through your body, you can feel a throbbing sensation (the pulse) by putting your fingers over one of your arteries at any point where the artery comes close to the surface of your skin, such as your wrist, neck, or upper arm.

Resting heart rate	
Age or fitness level	Beats per minute (bpm)
Babies to age 1:	100–160
Children ages 1 to 10:	60–140
Children age 10+ and adults:	60–100
Well-conditioned athletes:	40–60

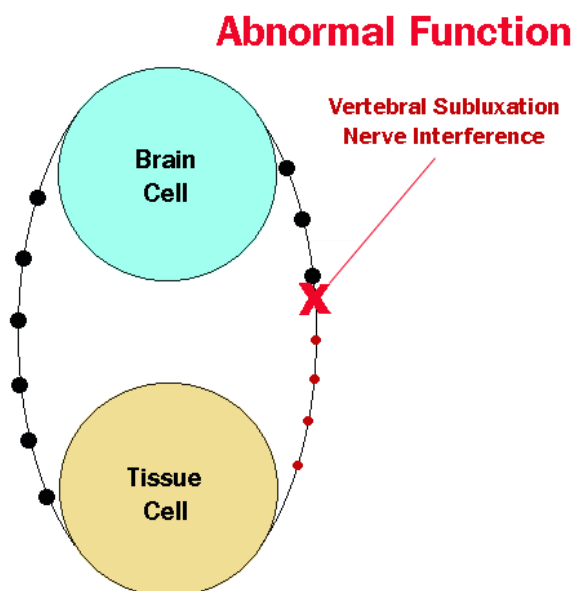
7. The nervous system

The nervous system is the major controlling, regulatory, and communicating system in the body. It is the center of all mental activity including thought, learning, and memory. Together with the endocrine system, the nervous system is responsible for regulating and maintaining homeostasis.

Through its receptors, the nervous system keeps us in touch with our environment, both external and internal. Like other systems in the body, the nervous system is composed of organs, principally the brain, spinal cord, nerves, and ganglia.

These, in turn, consist of various tissues, including nerve, blood, and connective tissue. Together these carry out the complex activities of the nervous system. The various activities of the nervous system can be grouped together as three general, overlapping functions:

- ♦ Sensory
- ♦ Integrative
- ♦ Motor



Millions of sensory receptors detect changes, called stimuli, which occur inside and outside the body.

They monitor such things as temperature, light, and sound from the external environment. Inside the body, the internal environment, receptors detect variations in pressure, pH, carbon dioxide concentration, and the levels of various electrolytes. All of this gathered information is called sensory input.

Sensory input is converted into electrical signals called nerve impulses that are transmitted to the brain. There the signals are brought together to create sensations, to produce thoughts, or to add to memory; Decisions are made each moment based on the sensory input. This is integration.

Based on the sensory input and integration, the nervous system responds by sending signals to muscles, causing them to contract, or to glands, causing them to produce secretions. Muscles and glands are called effectors because they cause an effect in response to directions from the nervous system. This is the motor output or motor function.

[illegible]

STUDY UNIT 3

MULTIPLE INJURIES, MULTIPLE INCIDENTS AND TRIAGE

LEARNING OUTCOMES

On completion of this study unit the student will....



- ♦ Define the meaning of triage.
- ♦ Explain the purpose of triage.
- ♦ Demonstrate knowledge of the various categories of triage and the application of treatment of each category.
- ♦ Demonstrate skills on the application of the methods to be used for triage.
- ♦ Apply the procedure when treatment for multiple casualty incidents needs to be effected.

1. Triage

DEFINITION

**SORTING MULTIPLE PATIENTS IN PRIORITIES
ACCORDING TO THE SERIOUSNESS OF THEIR
INJURIES AND CONDITIONS.**

1.1 Purpose of TRIAGE

The aim is to 'do the most for the most'. You divide patients into categories.

Casualties may be given colored triage label.

1.2 Triage categories

Category	Definition	Colour	Treatment	Example
P1	Life-threatening	Red	Immediate	Tension pneumothorax
P2	Urgent	Yellow	Urgent	Fractured femur
P3	Minor	Green	Delayed	Sprained ankle
P4	Dead	White		

REMEMBER.....

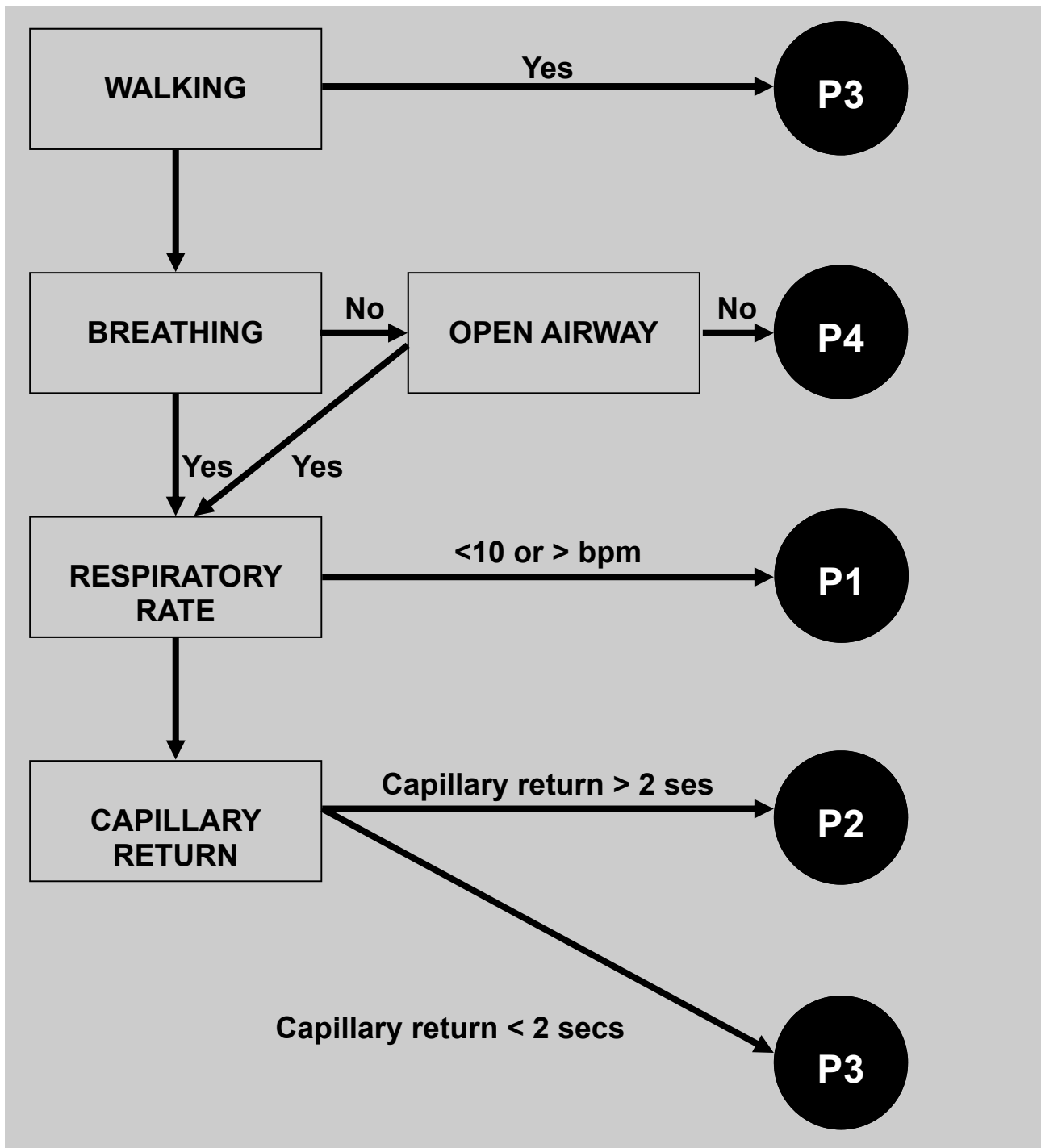
Always distinguish between conscious and unconscious patients.

1.3 Method of using triage

Triage can be performed rapidly by assessing.....

- ♦ Ability to walk
- ♦ Airway
- ♦ Respiratory rate
- ♦ Pulse rate or capillary return

See attached diagram



2. Multiple Casualty Incidents

TAKE NOTE

MULTIPLE CASUALTY INCIDENTS NEVER FOLLOW SET RULES.... They never go by the book.

REMEMBER.....

THE GOLDEN HOUR!!! TIME IS LIFE

TAKE NOTE

The principle immediate life threatening conditions are. **Respiratory arrest and Severe Bleeding.**

REMEMBER.....

Life has precedence over limb!!! First save the LIFE then the LIMB!!!!

2.1 Procedure

- ◆ Check the time.
- ◆ Use safety, gloves and mouthpiece.
- ◆ Communicate with all role players.

- ⇒ Offer assistance.
- ⇒ Incident history, mechanism of **injury**, clues.
- ⇒ Get help immediately! Call **ALS!**
- ⇒ Involve bystanders, other patients.

- ◆ Visually sort patients between obviously conscious and obviously unconscious.

- ◆ Conscious patients that bleed severely need immediate attention!

- ⇒ Stop severe bleeding by direct pressure and elevation - use patient, bystanders, etc. and position patient correctly.
- ⇒ Reassure and calm other patients and try to use them for assistance.

- ◆ Unconscious (or those who seem to be unconscious)

- ⇒ Stop bleeding
- ⇒ Check for response, determine if actually unconscious
- ⇒ Check ABC

- ☐ **A = airway, open (beware of neck injuries), finger sweep.**
- ☐ **B = breathing, look, listen and feel for 10 seconds, RB if necessary.**
- ☐ **C = circulation, check for 10 seconds, CPR if necessary, first stop bleeding**

- ◆ If ABC is in order, stop bleeding

- ⇒ Neck injuries, apply cervical collar
- ⇒ Position correctly

- ◆ If possible, do secondary survey on all patients and treat accordingly (if possible)

- ◆ Priority of treatment

- ☐ **Severe bleeding**
- ☐ **Shock**
- ☐ **Severe fractures**
- ☐ **Convulsions**
- ☐ **Minor injuries and conditions**

- ◆ Position patients
- ◆ Cover patient
- ◆ Monitor each patient's vital signs regularly

- ⇒ Airway
- ⇒ Breathing
- ⇒ Circulation
- ⇒ Temperature

- ◆ Record keeping
- ◆ Hand-over and report to ALS



[illegible]

STUDY UNIT 4

THE PRIMARY SURVEY

LEARNING OUTCOMES

On completion of this study unit the student will....



- ♦ Define the term primary survey.
- ♦ Identify the conditions of a patient that needs immediate attention.
- ♦ Identify and remove any element that can contribute to the emergency scene not being safe.
- ♦ Apply the correct procedure during the process of a primary survey.

1. Definition

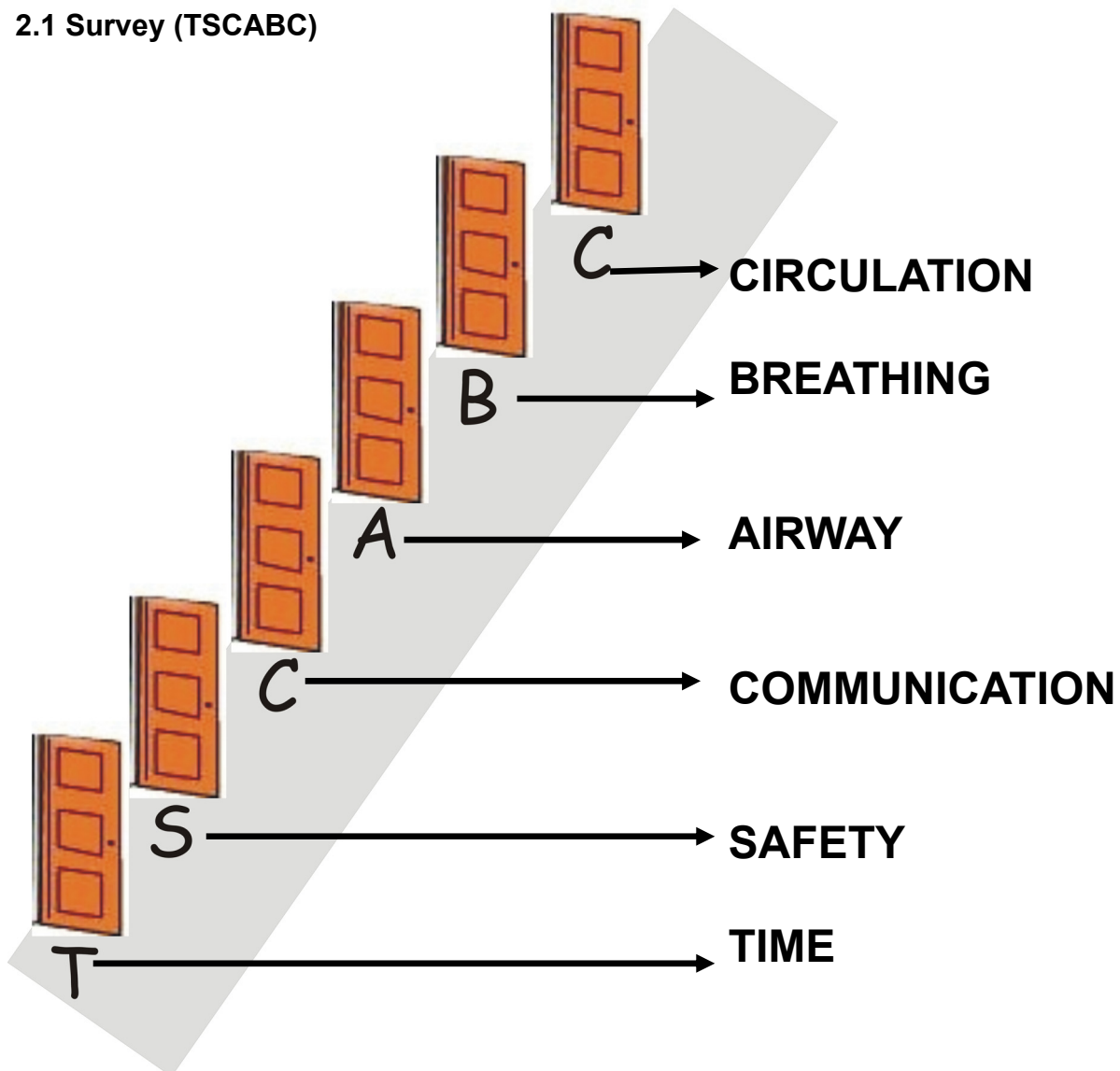
DEFINITION

The Primary Survey is a **process** of definitive steps to **critically survey** the scene, manage it, and to accurately **assess the patient** for acute life threatening conditions in order to treat them immediately.

2. The Process

TAKE NOTE**The ABC of FIRST AID**

2.1 Survey (TSCABC)



2.2 Diagnoses

REMEMBER.....

The four acute life threatening conditions:

- ♦ **Airway is obstructed.**
- ♦ **There is no breathing.**
- ♦ **There is not a pulse.**
- ♦ **Severe bleeding.**

2.3 Treatment

- ♦ Open the airway
- ♦ Rescue Breathing
- ♦ CPR
- ♦ Stopping severe bleeding

3. Time=T

- ♦ When did the incident happen? (lapsed time and duration of condition)
- ♦ Record the time that you have arrived on the scene.

4. Safety (as well as scene approach and management)=S

- ♦ Secure the scene and environment. Remove all possible Hazards.
- ♦ Secure your own safety. Use protective devices. I.e. Gloves and CPR mouthpiece.
- ♦ Secure the safety of bystanders.
- ♦ Secure the safety of your patient. Either remove your patient with care or remove the elements that endanger the patient.

5. Communication=C

TAKE NOTE

Provide the relevant information and seek relevant information.

- ◆ Introduce yourself, offer assistance and obtain consent for treatment.
- ◆ Obtain short facts from the patient and bystanders concerning the history of incident, mechanism of injury and the environment.
- ◆ Obtain information concerning the patient's general appearance, posture, condition and behavior by taking the following factors in mind:
 - ◆ State of consciousness
 - ⇒ Conscious (awake), Semi-conscious (Disorientated, confused or intoxicated)
 - ⇒ Unconscious
 - ◆ Patient's position.



**Beware of possible neck injuries
(mechanism of injury)**

- ⇒ Behavior and degree of distress.
- ⇒ Severe visible external bleeding.
- ◆ Assess the response of the patient
 - ⇒ Talk to the patient.
 - ⇒ Talk to the patient and touch the patient.
 - ⇒ Talk, touch and tap the patient.
- ◆ Obtain help from bystanders by shouting and calling for it.

Primary vital signs (ABC)

1. Conscious compared to unconscious patients
 - ⇒ Conscious - ABC is OK!
 - ⇒ Unconscious - ABC must be checked
2. Four conditions that need immediate attention (ABC)
 - ⇒ Obstructed Airway (A)
 - ⇒ Absence of Breathing (B)
 - ⇒ Absence of Circulation (pulse) (C)
 - ⇒ Severe bleeding

6. Airway open

- ◆ Your tongue is the main cause of airway obstruction in an unconscious patient.
- ◆ Other causes like constricting clothing, ties, position of patient and head.
- ◆ Maintain opening of the airway.

METHOD OF OPENING THE AIRWAY

1. head-tilt-chin-lift method.
2. jaw-thrust method.
3. finger sweep only if something is visible and can be reached with fingers.

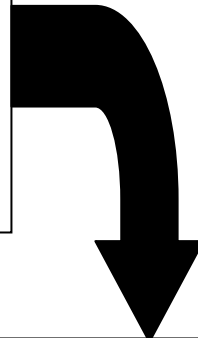
**IMPORTANT****PROTECT**

- ◆ The patient from swallowing his/her tongue.
- ◆ The patient if he/she are vomiting and or an excess of blood.
- ◆ The patient from rain and excess of water.
- ◆ The patient from smoke and gasses.

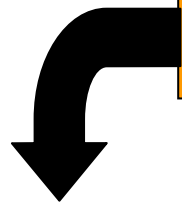
7. Breathing (B) (10 seconds, up to 30 seconds in hypothermia, certain overdoses, near-drowning)

TEST THE PATIENT BY:

1. Looking at movement of the chest.
2. Listen at breathing sound.
3. Feel air moving with holding your hand against the cheek of the patient.



IN THE EVENT OF NO BREATHING



1. Activate Emergency Services by phoning the following number:
 - ♦ 10177 or 112
2. Give two breaths.

8. Circulation=(C)

- ♦ Determine the presence of a pulse.
 - ⇒ Adults and children-carotid
 - ⇒ Infants-brachial

BLEEDING

- ♦ Always Check For Severe Bleeding And Stop Immediately!
- ♦ (Decide for yourself when this is necessary. i.e. before ABC)

TAKE NOTE

The following points are not part of the primary survey, but are things which should always be considered with all patients.

Other sources of help

- ♦ Fire Department
- ♦ Police
- ♦ Traffic

Positioning of the patient

- ♦ Recovery or coma, lateral, fowler, semi-fowler or shock
- ♦ Cover patient if necessary
- ♦ Regular Monitoring and recording

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STUDY UNIT 5

THE SECONDARY SURVEY

LEARNING OUTCOMES

On completion of this study unit the student will....



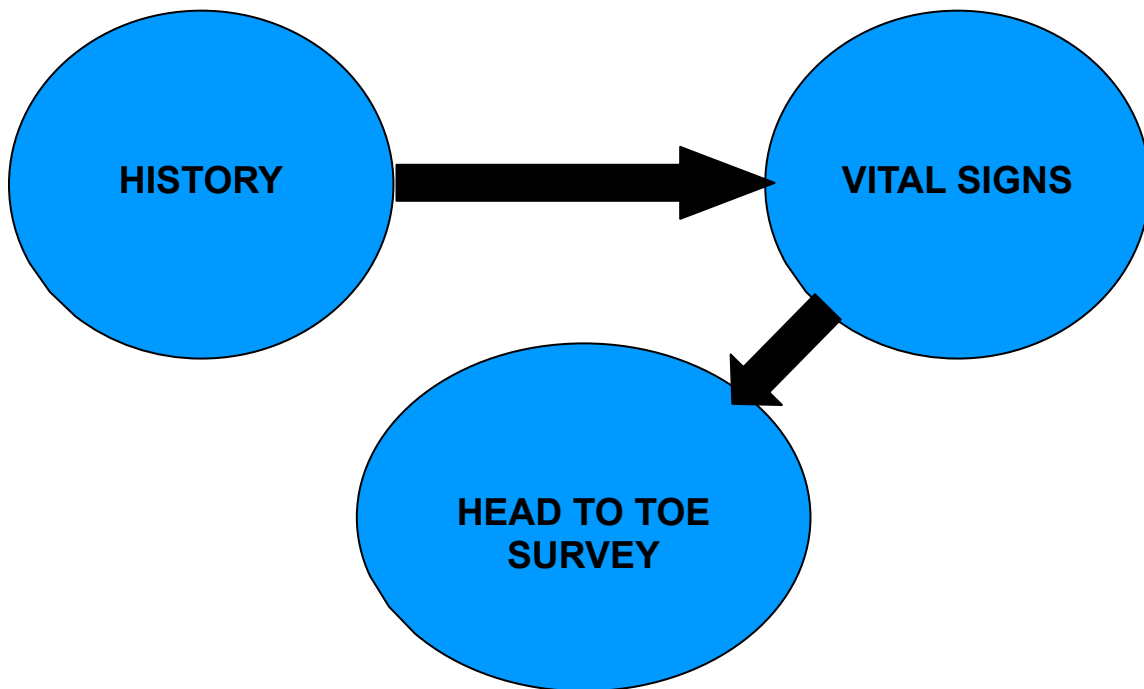
- ♦ Define the term secondary survey.
- ♦ Demonstrate knowledge of the elements of a secondary survey.
- ♦ Apply the correct procedure during the process of a secondary survey.

1. Definition

DEFINITION

To identify all other conditions that are not acutely life threatening but that may become life threatening or serious if not treated.

2. Elements of the secondary survey



2.1 Comprehensive history of the patient

Observation of the following factors is important.

- ♦ Specific signs and symptoms.
- ♦ Mechanism of injury.
- ♦ SOS, Medic Alert tokens.
- ♦ Diseases, allergies and medications used.

2.2 Vital signs

Determine the level of consciousness. Is the patient.....?

- ♦ Conscious
- ♦ Disorientated or confused
- ♦ Drowsy or sleepy
- ♦ Stupor
- ♦ Coma
- ♦ Death

Determine breathing.

- ◆ Type of breathing. Is it normal and easy, weak, forced, wheezing (whistling), snoring or gurgling.
- ◆ Heart rate. Beats per minute (normal = 12 - 20 per minute).
- ◆ The rhythm of breathing.
- ◆ The depth of breaths. Is it normal, shallow or deep?
- ◆ Determine whether there is a pulse.
- ◆ Pulse rate. Beats per minute (normal = 60 - 90 per minute).
- ◆ The rhythm of pulse. Is it regular?
- ◆ Strength. Is it normal, shallow or deep?

Determine the temperature of the patient.

- ◆ Is the temperature of the patient normal, hot or cold?

2.3 Head to toe surveys (adults)

2.3.1 Method.

Visually inspection: This is done by only looking at the patient.	THIS CAN ONLY BE ACHIEVED WHEN YOU USE ALL POSSIBLE SENSES.
Listening	
Feeling	
Smelling	

2.3.2 Physical assessment.



When you do a physical assessment you must distinguish between conscious and unconscious patients.

HEAD:

Observe the following important issues:

- ♦ General appearance, position and shape.
- ♦ Scalp, skull.
- ♦ Symmetry of face.
- ♦ Skin colour. Is it pale, blue, red or yellow (mucous membranes, tongue).
- ♦ Skin moisture.
- ♦ Eyes, eye sockets.
- ♦ Pupils, P.E.R.L.
- ♦ Nose.
- ♦ Upper jaw and cheek-bones.
- ♦ Lips, mouth, teeth and tongue.
- ♦ Jaw-bone.
- ♦ Ears, mastoid processes.

NECK AND THROAT

- ♦ Look at injuries on the neck including but limited to bleeding and wind pipe.

COLLOR BONE AND HEAD OF HUMEROUS

- ♦ Possible fracture

CHEST

- ♦ Symmetry.
- ♦ Breast bone.
- ♦ Ribs, downwards and sideways pressure.
- ♦ Breathing and chest movement.
- ♦ Whistling or sucking sounds from the chest.

ABDOMEN

- ♦ Quadrants.
- ♦ Visible signs.
- ♦ Elasticity.
- ♦ Needle marks.

PELVIS

- ♦ Downwards and sideways pressure on pelvic bones and pubis.
- ♦ Uncontrolled bowel and urinary discharge.
- ♦ Erection.
- ♦ Femur heads.

UPPER AND LOWER LIMBS

- ♦ Natural ability to move, paralysis and weakness. Compare with opposite limb.
- ♦ Deformity.
- ♦ Dislocations.
- ♦ Fractures, mobility and sensation swelling
- ♦ Wounds and bleeding.
- ♦ Abduction and adduction.
- ♦ Flexion and extension.
- ♦ Rotation.
- ♦ Comparative length, colour and temperature.
- ♦ Distal pulse, capillary refill.

BACK, SPINE AND SHOULDER BLADE

- ♦ Possible fractures.

2.4 Head to toe survey for children and infants

- ♦ The physical examination of children and infants should be performed from the toes to the head.

3. Positioning of patient

Place patient in one of the following positions:

- ♦ Shock position
- ♦ Coma or recovery and lateral position
- ♦ Fowler and Semi-Fowler position





4. Follow up care, monitoring and recording

- ♦ Regular monitoring, every 5 minutes
- ♦ Check the vital signs
- ♦ Keep records

Hand-over and report to advanced care

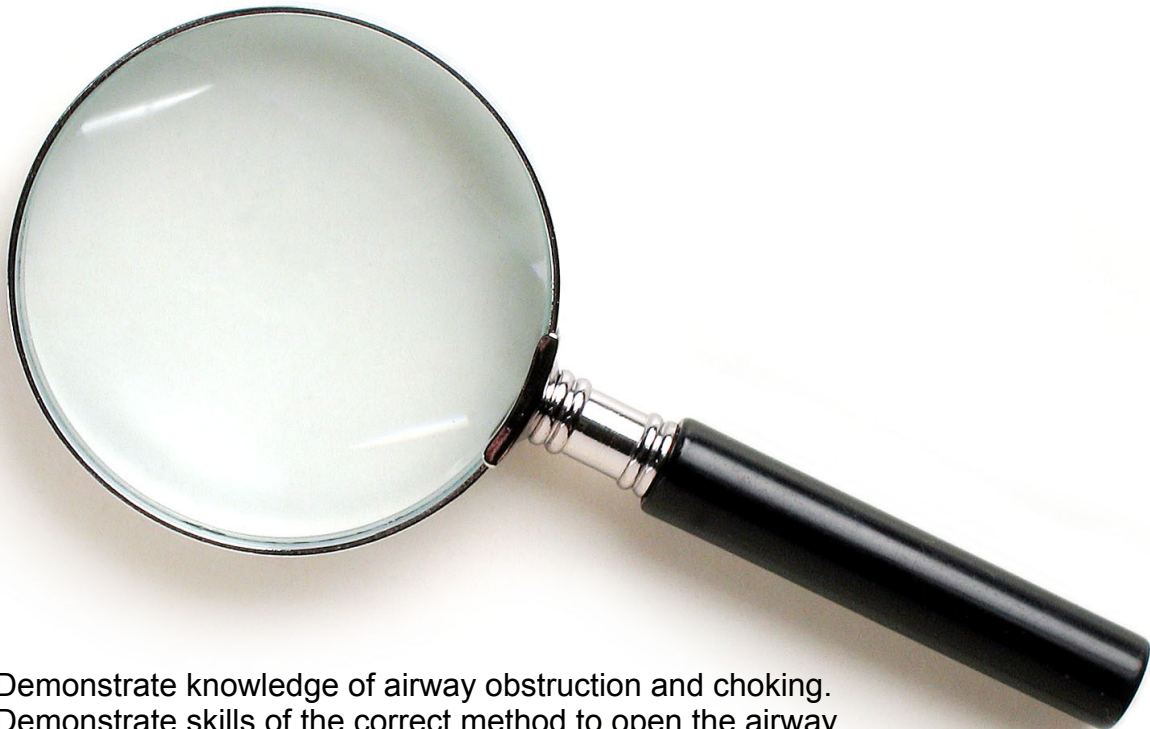
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STUDY UNIT 6

AIRWAY OBSTRUCTION AND RESCUE BREATHING

LEARNING OUTCOMES

On completion of this study unit the student will....



- ♦ Demonstrate knowledge of airway obstruction and choking.
- ♦ Demonstrate skills of the correct method to open the airway.
- ♦ Distinguish the difference between partial obstruction and total obstruction.
- ♦ Demonstrate knowledge of the signs and symptoms of partial obstruction.
- ♦ Demonstrate skills for the correct treatment of partial obstruction.
- ♦ Demonstrate knowledge of the signs and symptoms of total obstruction.
- ♦ Demonstrate skills for the correct treatment of total obstruction.
- ♦ Demonstrate skills of the application of the various methods that can be used to deal with choking of a:
 - ⇒ Adult
 - ⇒ Infant or baby

1. Airway obstruction and choking (Dealing with the "A" of ABC)



DEFINITION

Obstruction and or choking is caused by an obstruction in the airway, normally caused by food or vomiting lodged in the airway or under the valve where air enters into the lungs.

Airway open

- ♦ Your tongue is the main cause of airway obstruction in an unconscious patient.
- ♦ Other causes like constricting clothing, ties, position of patient and head.
- ♦ Maintain opening of the airway.

METHOD OF OPENING THE AIRWAY

1. head-tilt-chin-lift method
2. jaw-thrust method
3. finger sweep only if something is visible and can be reached with fingers



IMPORTANT



PROTECT

- ♦ The patient from swallowing his/her tongue.
- ♦ The patient if he/she are vomiting and or an excess of blood.
- ♦ The patient from rain and excess of water.
- ♦ The patient from smoke and gasses.

Simulated universal distress signal.

Universal sign
for choking

2. Partial Obstruction

The airway is partially obstructed by any of the following things:

- ♦ Food
- ♦ Any small object stuck in the throat of the patient
- ♦ Injuries, vomiting

Signs of partial obstruction.

- ♦ The patient is able to make sounds and be able to cough
- ♦ Patient will clutch his throat and have difficulty in breathing
- ♦ He may turn red in the face and become anxious
- ♦ The veins will protrude
- ♦ Whistling sounds may be heard

2.1 Treatment

- ♦ Remain calm and re-assure the patient.
- ♦ Ask for assistance and request a bystander to call for an ambulance.
- ♦ Encourage the patient to cough. This can possibly dislodge the obstruction.
- ♦ If the patient can cough do not intervene.

3. Total obstruction

Three specific signs of a total airway obstruction are.

- ♦ The patient cannot speak
- ♦ The patient cannot cough
- ♦ The patient cannot breathe

3.1 Treatment

Conscious adult patient (witnessed choking)

IMPORTANT

Get advanced help before you do anything!!

- ♦ Remain calm and re-assure the patient.
- ♦ Ask for assistance and request a bystander to call for an ambulance.
- ♦ Encourage the patient to cough. This can possibly dislodge the obstruction.
- ♦ If the patient can cough do not intervene.
- ♦ Instruct the patient to bend forward with the head lower than the chest.

TAKE NOTE

Small children and babies can be placed across your knee with the head down.

- ♦ Apply a few hard thumps between his/her shoulders with the heel of your hand.
- ♦ Do this a few times.
- ♦ Inspect the inside of the mouth. Either you or the patient should do a finger sweep of the mouth and try to dislodge the blockage.

CRITICAL**SHOULD THE CHOKING PERSIST:
PROCEED WITH THE HEIMLICH MANOEUVRE****PROCEDURE TO APPLY HEIMLICH MANOEUVRE****When Standing:**

1. Place yourself behind the patient.
2. Place your arms around his body.
3. Clench the fist of your one hand and place your hand over the fist.
4. Place your fist on the patient's upper abdomen under the diaphragm.
5. Gently jerk your hands inwards and upwards.



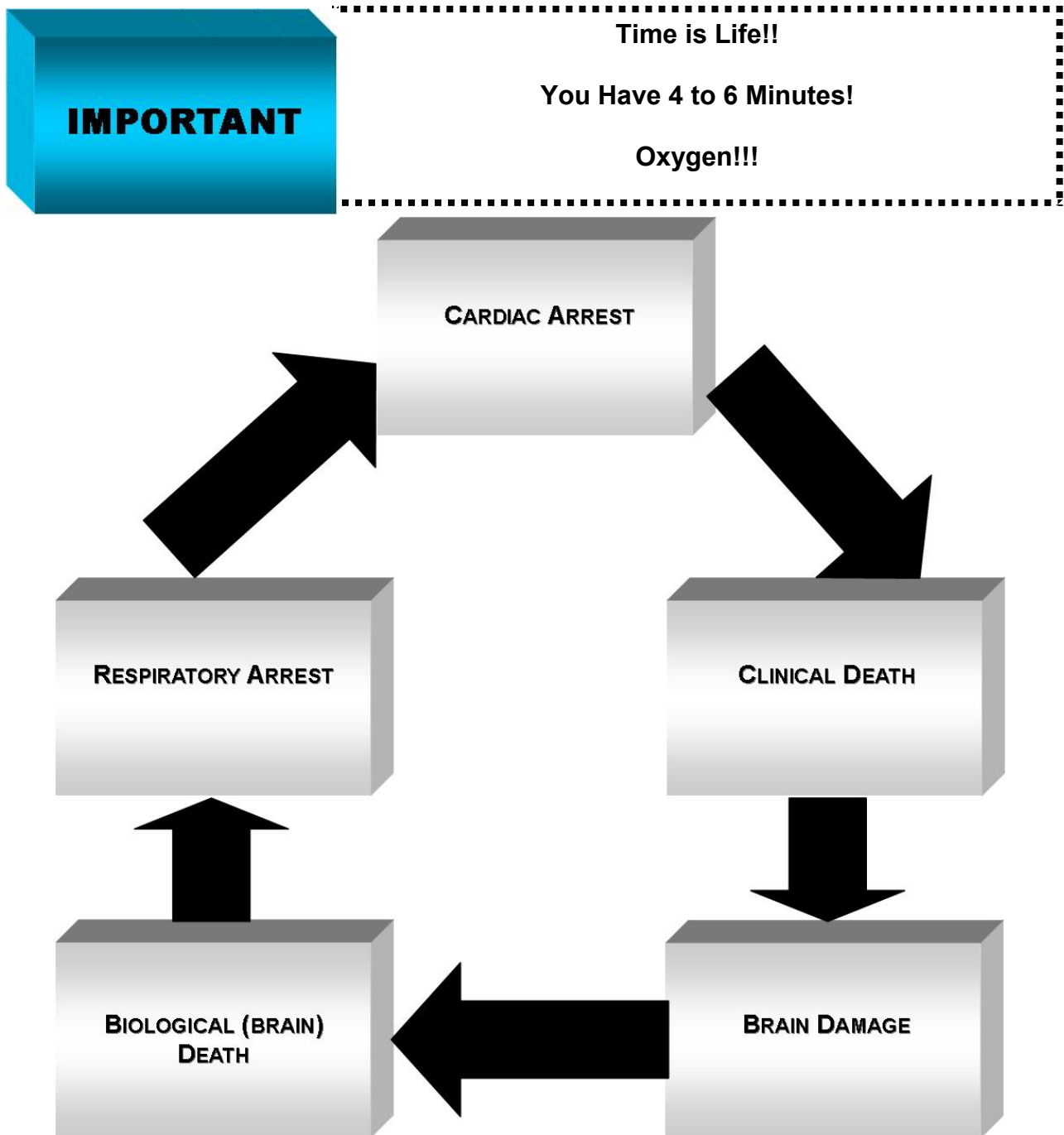
Infant or baby**Step 1: Figure out what's causing the problem**

- ♦ If your baby is suddenly unable to cry or cough she probably has something blocking her airway, and you will need to help her get it out.

Step 2: Give back blows and chest thrusts

If your baby can't clear her airway on her own and you believe something is trapped there, carefully position her face down on your forearm, with your hand supporting her head and neck, and rest the arm holding the baby on your thigh. Support her so that her head is lower than the rest of her body. Then give her five back blows between her shoulder blades with the heel of your hand to try to dislodge the object. Next, place your free hand (the one that had been delivering the back blows) on the back of her head and along her spine and carefully turn her over supporting the head and neck.



4. Rescue Breathing - RB (Dealing with the "B" of ABC)

4.1 Causes of respiratory arrest and airway obstruction:

- ♦ Lack of oxygen
- ♦ Foreign objects, fluids, choking, suffocation
- ♦ Injuries: chest, brain, spine
- ♦ Diseases: asthma, emphysema, bronchitis, croup, epiglottitis

[illegible]

STUDY UNIT 7

ARTIFICIAL RESPIRATION (Mouth to mouth ventilation)

LEARNING OUTCOMES

On completion of this study unit the student will....



- ◆ Demonstrate knowledge of the causes of respiratory arrest and airway obstruction.
- ◆ Demonstrate knowledge of the meaning of artificial respiration.
- ◆ Demonstrate knowledge and skills in the importance of using:
 - ⇒ Protective gloves (latex)
 - ⇒ Protective equipment (mouth pieces)
- ◆ Demonstrate the correct knowledge during the event of mouth to mouth ventilation.
- ◆ Demonstrate the necessary skills in the application of artificial respiration methods.

IMPORTANT

Time is Life!!

You Have 4 to 6 Minutes!

Oxygen!!!

DEFINITION

Artificial respiration is a technique for providing air for a person who is not **breathing** on their own but whose heart is still beating. The provider breathes into the other person's lungs, preferably with assistance of a barrier device.



1. Open the Airway

METHOD OF OPENING THE AIRWAY

1. head-tilt-chin-lift method
2. jaw-thrust method

1.1 Head tilt chin lift

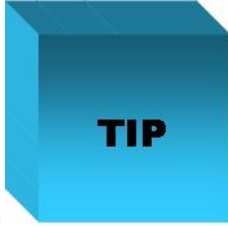
This is probably the most common method to apply.



- ◆ Do not perform if neck injury suspected.
- ◆ Use **Jaw Thrust** instead
- ◆ Avoid closing mouth or compressing chin soft-tissue it can obstruct the airway.

PROCEDURE:

One hand on the forehead to tilt head back

FOR INFANTS	
1.	Place head in neutral position (sniffing position)   Do not overextend head and neck

FOR CHILDREN AND ADULTS

Head and neck slightly Extended

The line from the chin to the jaw must form an angle perpendicular to the floor.

Use the other hand's fingers under the bony part of the chin.

- ♦ Do not use your thumb to lift the chin.
- ♦ Lift the mandible upward and outward.







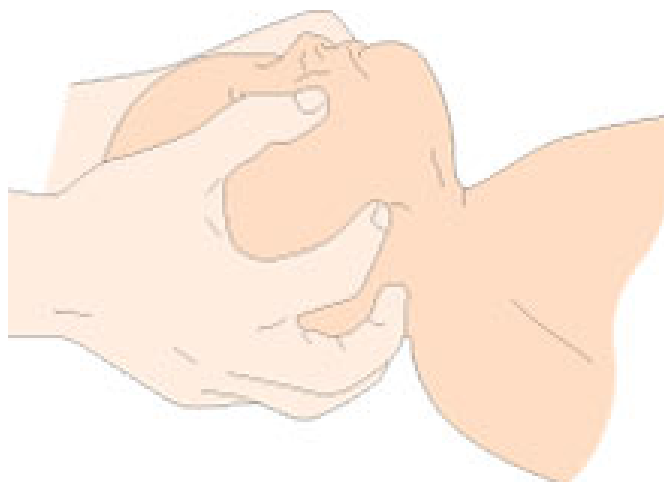
1.2 Jaw Thrust

This is done by opening the airway in suspected Cervical Spine Injury.

Assistants must also perform other activities such as cervical spine immobilization and ventilation

PROCEDURE

- ◆ Place 2 to 3 fingers under each side of the lower jaw angle.
- ◆ Lift the jaw upwards and outwards.

INFANTS**CHILDREN AND ADULTS**

[illegible]

STUDY UNIT 8

CPR (Cardio Pulmonary Resuscitation)

LEARNING OUTCOMES

On completion of this study unit the student will....



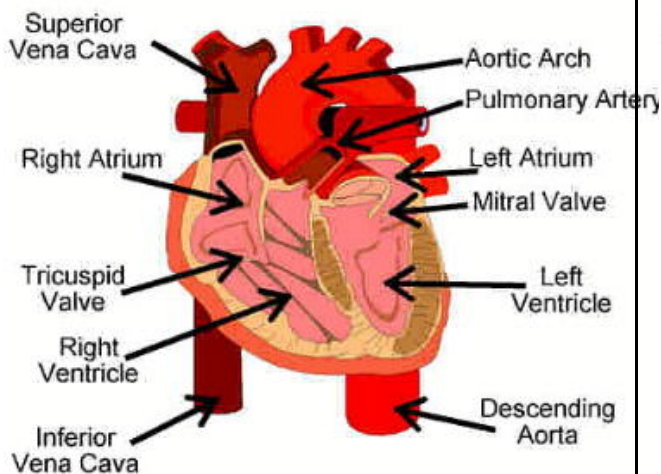
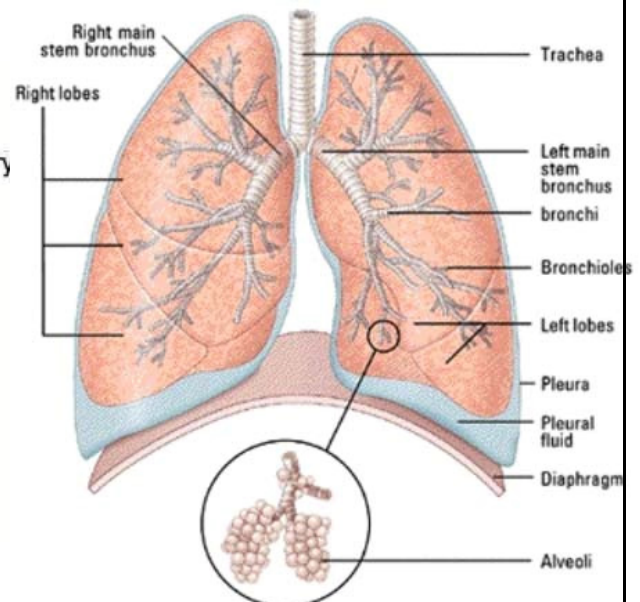
- ♦ Define and demonstrate knowledge of CPR.
- ♦ Demonstrate knowledge and skill of the procedure to clear the airway in the event of CPR.
- ♦ Demonstrate knowledge and skill of the procedure to breath for a patient in the event of CPR.
- ♦ Demonstrate knowledge and skill of the procedure to restore the blood circulation of a patient in the vent of CPR.
- ♦ Demonstrate skill in the application of CPR on
 - ⇒ A child
 - ⇒ A adult
 - ⇒ A baby

CPR (Cardio Pulmonary Resuscitation)

DEFINITION

Cardiopulmonary resuscitation (CPR) is a lifesaving technique useful in many emergencies, including heart attack or near drowning, in which someone's breathing or heartbeat has stopped. CPR involves a combination of mouth-to-mouth rescue breathing and chest compression that keeps oxygenated blood flowing to the brain and other vital organs until more definitive medical treatment can restore a normal heart rhythm.

When the heart stops, the absence of oxygenated blood can cause irreparable brain damage in only a few minutes. Death will occur within 8 to 10 minutes. Time is critical when you're helping an unconscious person who isn't breathing.

C=CARDIAC (HEART)**P=PULMANORY (LUNGS)****R=RESUCITATION (BRING BACK TO LIFE)**

IMPORTANT**Before you begin:**

Assess the situation before starting CPR:

- ♦ Is the person conscious or unconscious?
- ♦ If the person appears unconscious, tap or shake his or her shoulder and ask loudly, "Are you OK?"
- ♦ If the person doesn't respond, call your local emergency number), or have someone else do it.

REMEMBER.....

If you're alone and the patient is an infant or a child age 1 to 8 who needs CPR, perform two minutes of CPR before calling for help.

1. The ABC's

A	AIRWAY. IS IT OPEN?
B	BREATHING. IS THE PATIENT BREATHING?
C	CIRCULATION. IS THE HEART BEATING?

Clear the airway





Look, listen and feel for breathing and pulse

PROCEDURE	
1.	Put the person on his back on a firm surface.
2.	Kneel next to the person's neck and shoulders.
3.	Open the person's airway using the head tilt-chin lift. Put your palm on the person's forehead and gently push down. Then with the other hand, gently lift the chin forward to open the airway.
4.	Check for normal breathing, taking no more than 10 seconds: Look for chest motion, listen for breath sounds, and feel for the person's breath on your cheek and ear. Do not consider gasping to be normal breathing. If the person isn't breathing normally or you aren't sure, begin mouth-to-mouth breathing.

Breathing. Breath for the person

Rescue breathing can be mouth-to-mouth breathing or mouth-to-nose breathing if the mouth is seriously injured or can't be opened.



4

Place your mouth over the patient's mouth and exhale.



5



PROCEDURE	
1.	With the airway open (using the head tilt-chin lift), pinch the nostrils shut for mouth-to-mouth breathing and cover the person's mouth with yours, making a seal.
2.	Prepare to give two breaths. Give the first rescue breath, lasting one second, and watch to see if the chest rises. If it does rise, give the second breath. If the chest doesn't rise, repeat the head tilt-chin lift and then give the second breath.
3.	Begin chest compressions; go to "CIRCULATION" below.

Circulation: Restore the blood circulation



PROCEDURE	
1.	Place the heel of one hand over the center of the person's chest, between the nipples. Place your other hand on top of the first hand. Keep your elbows straight and position your shoulders directly above your hands.
2.	Use your upper body weight (not just your arms) as you push straight down on (compress) the chest 1 1/2 to 2 inches. Push hard and push fast, give two compressions per second, or about 100 compressions per minute.
3.	After 30 compressions, tilt the head back and lift the chin up to open the airway. Prepare to give two rescue breaths. Pinch the nose shut and breathe into the mouth for one second. If the chest rises, give a second rescue breath. If the chest doesn't rise, repeat the head tilt-chin lift and then give the second rescue breath. That's one cycle. If someone else is available, ask that person to give two breaths after you do 30 compressions.
4.	Continue CPR until there are signs of movement or until emergency medical personnel take over.

2. CPR

2.1 CPR on a child/adult

The procedure for giving CPR to a child age 1 through 8 is essentially the same as that of an adult. The differences are as follows:

1. Perform five cycles of compressions and breaths on the child. This should take about two minutes before calling for advance emergency care.



2. Use only one hand to perform heart compressions.
3. Breath more gently
4. Use the same compression/breath rate as is used for adults: 30 compressions followed by two breaths.

This is one cycle. Following the two breaths, immediately begin the next cycle of compressions and breaths. Continue until patient moves or help arrives.

2.2 CPR on a baby

Most cardiac arrests in infants occur from lack of oxygen, such as from drowning or choking. If you know the infant has an airway obstruction, perform first aid for choking. If you don't know why the infant isn't breathing, perform CPR.

To begin, assess the situation. Stroke the baby and watch for a response, such as movement, but **DON'T SHAKE THE CHILD**

If there is response, follow **the ABC procedures below** and time call for help as follows:

- | | |
|----|--|
| 1. | If you are the only first aider and CPR is needed, do CPR for two minutes, about five cycles before you call for help. |
| 2. | If another person is nearby and available, have that person call for help. |

Clean the airway:

- ♦ Place the baby on his or her back on firm, flat surface, such as a table. The floor or ground also will do.
- ♦ Gently tip the head back by lifting the chin with one hand and pushing down on the forehead with the other hand.
- ♦ In no more than 10 seconds, put your ear near the baby's mouth and check for breathing: Look for chest motion, listen for breath sounds and feel for breath on your cheek and ear.



If the infant is not breathing, begin with mouth to mouth breathing immediately.

Breathing: Breath for the infant

- ♦ Cover the baby's mouth and nose with your mouth or mouth piece.
- ♦ Prepare to give two rescue breaths. Use the strength of your cheeks to deliver gentle puffs of air (instead of deep breaths from your lungs) to slowly breathe into the baby's mouth one time, taking one second for the breath. Watch to see if the baby's chest rises. If it does, give a second rescue breath. If the chest does not rise, repeat the head-tilt-chin lift and give the second breath.
- ♦ If the chest still doesn't rise, examine the mouth to make sure no foreign material is inside. If the object is seen, sweep it out with your finger. If the airway seems blocked, perform first aid for a choking infant.
- ♦ Begin chest compressions

**Circulation: Restore blood circulation**

- ♦ Imagine a horizontal line drawn between the baby's nipples. Place two fingers of one hand just below this line, in the center of the chest.
- ♦ Gently compress the chest to about one-third to one half the depth of the chest.
- ♦ Count aloud as you pump in a fairly rapid rhythm. You should pump at a rate of about 100 times a minute.
- ♦ Give two breaths after every 30 chest compressions.
- ♦ Perform CPR for about two minutes before calling for help unless someone else can make the call.
- ♦ Continue CPR until you see signs of life or until emergency care relieves you.



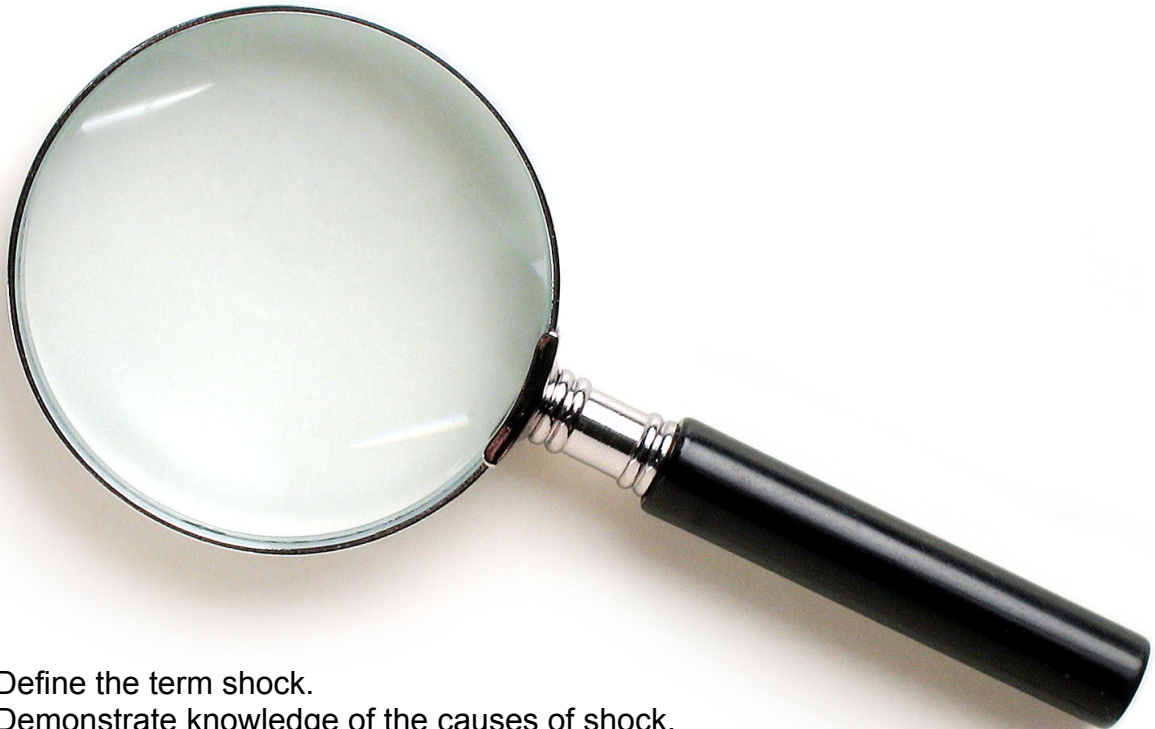
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STUDY UNIT 9

SHOCK

LEARNING OUTCOMES

On completion of this study unit the student will....



- ♦ Define the term shock.
- ♦ Demonstrate knowledge of the causes of shock.
- ♦ Demonstrate knowledge of the different types of shock.
- ♦ Demonstrate knowledge of the signs and symptoms of shock during the early stages and later stages of shock.
- ♦ Demonstrate skills in the treatment of shock.

1. Shock

DEFINITION

Shock is a medical emergency caused by a chain of events that result in insufficient circulation of oxygen rich blood to tissues and organs.

1.1 Causes of shock

- ♦ Loss of fluid (blood, plasma and fluids)
- ♦ Pump failure (Heart attack, heart failure, heart injury)
- ♦ Dilated (widened) blood vessels (Spinal cord injuries, poisoning, overdoses, pain).

1.2 Types of shock

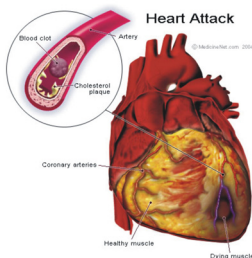
HYPOVOLEMIC SHOCK



♦ Hypovolaemic Shock

Too little blood to carry sufficient oxygen deprives body of oxygen.

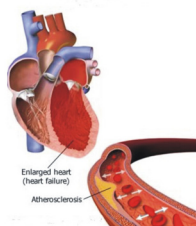
CARDIOGENIC SHOCK



♦ Cardiogenic Shock

Heart not able to effectively circulate oxygen rich blood, deprives body of oxygen.

NEUROGENIC SHOCK



♦ Neurogenic Shock

Blood vessels dilated resulting in too little blood for enlarged vessels, deprives body of oxygen.

2. Signs and symptoms of shock

2.1 Early signs

- ◆ Listlessness, apprehensive, anxiousness, scared, pale, cool and moist skin.

2.2 Later signs (Actual start of shock process)

- ◆ Still pale, cool and moist skin.
- ◆ Perspiration.
- ◆ Nausea and possible vomiting.
- ◆ Thirst and dry lips.
- ◆ Eyes dull and expressionless.
- ◆ Pupils dilated, slow reacting.
- ◆ Lowered level of consciousness, may become unconscious.
- ◆ Weakness.
- ◆ Breathing at first slow and deep, later rapid and shallow.
- ◆ Pulse rapid and weak.
- ◆ Shivering.

2.3 Treatment

- ◆ TSCABC! (time, safety, communication, airway, breathing and circulation)
- ◆ Control severe bleeding. (If necessary before TSCABC!)
- ◆ Lift legs 30cm (shock position) (except with injuries to the neck, back, head, abdomen, pelvis, leg fractures and especially with heart attack)
- ◆ Recovery position if unconscious (carefully observe patient for vomiting and aspiration)
- ◆ Cover patient, maintain body temperature. (space blanket)
- ◆ Nothing per mouth! Wet lips if very thirsty.
- ◆ Monitor vital signs and keep record.



IMPORTANT

If breathing becomes weak, or the patient becomes unconscious or begins to vomit, place him/her in the recovery position.

**If the patient stops breathing or the heart stop beating,
immediately start with CPR.**



3. Recovery Position



[illegible]

STUDY UNIT 10

BLEEDING

LEARNING OUTCOMES

On completion of this study unit the student will....



- ♦ Define:
 - ⇒ Arterial bleeding
 - ⇒ Venous bleeding
- ♦ Demonstrate knowledge of the classification of the categories of bleeding.
- ♦ Demonstrate knowledge of the signs and symptoms of severe bleeding.
- ♦ Demonstrate skill to control bleeding.

1. Definitions**DEFINITION****1.1 Arterial bleeding**

Bright red colour - oxygen rich, spurts out, can be fatal very quickly.

1.2 Venous bleeding

Dark red colour (maroon) - oxygen poor, flows out.

2. Classification**2.1 External bleeding**

You can see external bleeding.

2.2 Internal Bleeding

Internal bleeding is most of the time not visible.

2.3 Subcutaneous bleeding.

Normally dark bruises



3. Signs and symptoms of severe bleeding

- ♦ Listless, apprehensive
- ♦ Skin pale, cool and moist
- ♦ Thirst and dry lips
- ♦ Eyes dull, pupils dilated and slow reacting
- ♦ Nausea and possible vomiting
- ♦ Breathing rapid, yawning and sighing. (air hunger)
- ♦ Anxiety
- ♦ Lowered level of consciousness, may become unconscious
- ♦ Pulse rapid and weak
- ♦ Depression
- ♦ Shivering

4. Control of natural bleeding

- ♦ Clotting
- ♦ Remember that blood vessels are elastic and that blood vessels is constricted



Blood pressure lowers during bleeding

5. External control and treatment.

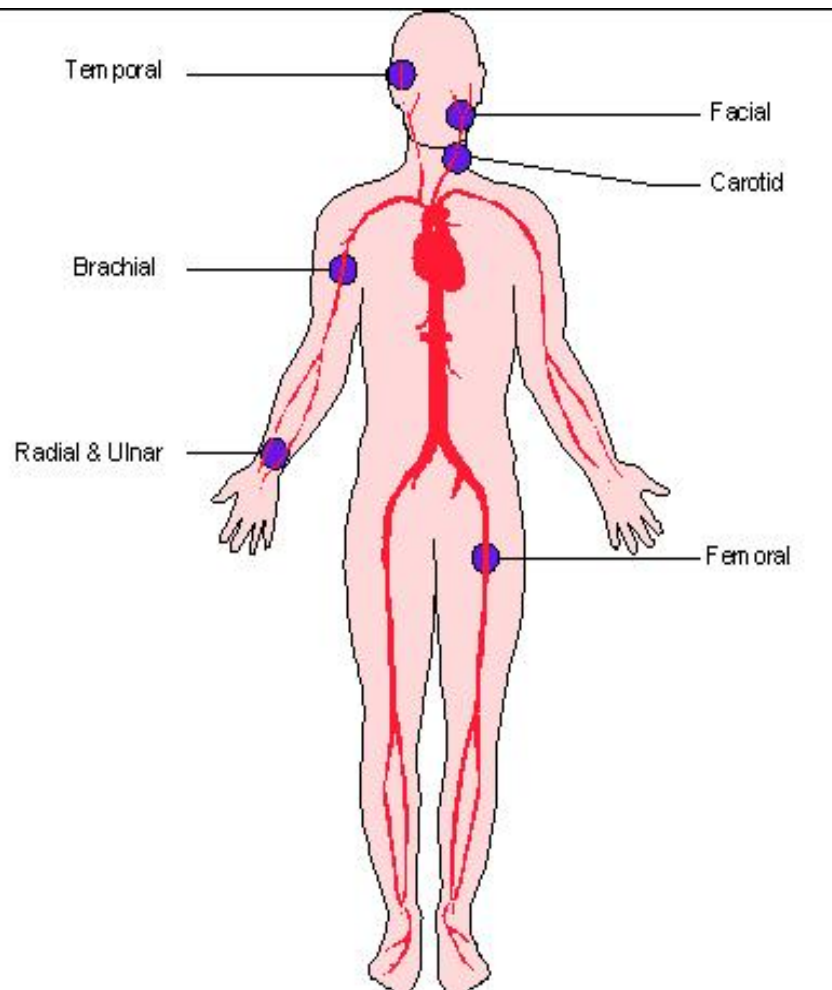
- ♦ TSCABC! (time, safety, communication, airway, breathing and circulation)
- ♦ If necessary, always remove (or cut away) clothing and jewellery before treating.

5.1 Severe external bleeding. Initial control.

- ♦ elevation
- ♦ direct pressure
- ♦ pressure points

5.2 Final treatment

- ♦ Pressure pads and pressure bandages
- ♦ Ring bandage. scalp and eye sockets

PRESSURE POINTS**6. Internal bleeding (signs, symptoms and treatment)****6.1 Stomach**

- ♦ Vomiting of blood, dark brown (like coffee grounds)

6.2 Lungs

- ♦ Coughing of blood, pink and frothy

6.3 Bladder

- ♦ Discoloration of the urine

6.4 Intestines

- ♦ Black stools

CRITICAL

Treating severe internal bleeding, is to treat for shock and hospitalise urgently

[illegible]

STUDY UNIT 11

UNCONSCIOUSNESS, HEAD AND BRAIN INJURIES

LEARNING OUTCOMES

On completion of this study unit the student will....



- ♦ Demonstrate knowledge of the levels of consciousness and unconsciousness.
- ♦ Distinguish between a stupor and coma.
- ♦ Demonstrate knowledge of the causes of unconsciousness.
- ♦ Demonstrate the skill to treat consciousness and unconsciousness patients.
- ♦ Demonstrate skill and knowledge to treat patients with head injuries.
- ♦ Demonstrate knowledge of the recovery position.
- ♦ Demonstrate skill of the procedure to place a patient in the recovery position.

1. Levels of consciousness and unconsciousness

- ♦ Normal
- ♦ Confused or disorientated
- ♦ Drowsy or sleepy
- ♦ Stupor
- ♦ Coma
- ♦ Death

2. Difference between stupor and coma

CRITERIA	COMA	STUPOR
CAN IT BE AWAKEN	NO	YES
IS THERE A CORNEAL REFLEX	NO	YES
IS THERE A PUPIL REACTION	NO	YES

3. Causes of unconsciousness

- ♦ Head and brain injuries
- ♦ Lack of oxygen
- ♦ Overdose, poisons
- ♦ Diseases
- ♦ Psychological reasons
- ♦ Fever convulsions
- ♦ Heat stroke

4. General treatment of unconscious patients**TSCABC! (time, safety, communication, airway, breathing and circulation)**

- ♦ Treat severe bleeding and fractures
 - ♦ Suspect shock and treat
 - ♦ Suspect neck and spinal injuries, cervical collar and spine board
 - ♦ Monitor the following regularly
- ⇒ LOC
- ⇒ breathing
- ⇒ pulse
- ⇒ temperature
- ⇒ beware of convulsions that may start, protect patient against further injuries
- ♦ Recovery position
 - ♦ Get ALS (Advance Life Support) if patient does not regain conscious after 5 minutes

5. Head injuries**5.1 Types of head injuries**

- ◆ Scalp injuries: abrasions, bruises and cuts and lacerations.
- ◆ Skull fractures and compression injuries.
- ◆ Brain injuries: concussion, lacerations, bruises and bleeding inside the skull and brain.

5.2 Signs and symptoms (almost the opposite of shock)

- ◆ History, Mechanism of injury and associated injuries.
- ◆ Level of consciousness may drop suddenly.
- ◆ Dizziness.
- ◆ Headache.
- ◆ Pupils unequal slow reacting and dilated.
- ◆ Squint eyes, seeing stars and spots.
- ◆ Slow breathing or difficult breathing.
- ◆ May stop breathing because of increased pressure inside the skull.
- ◆ Pulse slow and strong.
- ◆ Blood or fluid from the ears or nose. (Cerebro-spinal fluid)
- ◆ Raccoon eyes.
- ◆ Discoloring of mastoid processes. (Battle sign)
- ◆ Paralysis or weakness of limbs.
- ◆ Unnatural attitude of limbs. (flexion and extension)
- ◆ Convulsions.
- ◆ Nausea and strong vomiting.
- ◆ Aggressive and combative behavior.

5.3 Treatment**Scalp injuries**

- ◆ TSCABC! - RB and CPR may be necessary.
- ◆ Check LOC regularly.
- ◆ Suspect skull fractures and brain injuries.
- ◆ Suspect neck injuries.
- ◆ Cervical collar and spine board if necessary.
- ◆ Ring bandage to control bleeding.
- ◆ Positioning.

- ☐ Semi-Fowler (conscious)
- ☐ recovery position (unconscious)

ALS! (Advance Life Support)

Skull fractures

- ♦ TSCABC! - RB and CPR may be necessary.
- ♦ Check LOC regularly.
- ♦ Suspect brain injuries.
- ♦ Suspect neck injuries.
- ♦ Cervical collar and spine board.
- ♦ Check for blood and fluid from nose and ears - never stop it! Just cover it lightly.
- ♦ Positioning.

- ☐ Semi-Fowler (conscious)
- ☐ Recovery position (unconscious)

- ♦ ALS! (Advance Life Support)

Brain injuries

- ♦ TSCABC! - RB and CPR may be necessary.
- ♦ Check LOC regularly.
- ♦ Suspect neck injuries.
- ♦ Cervical collar and spine board.
- ♦ Check for blood and fluid from nose and ears - never stop it! Just cover it lightly.
- ♦ Positioning.

- ☐ semi-Fowler (conscious)
- ☐ recovery position (unconscious)

- ♦ ALS! (Advance Life Support)

6. The recovery position

If the person is unconscious, there is a safe position to put them in which allows them to breathe easily and stops them choking on any vomit.

6.1 Procedure for placing patient in the recovery position.

Once you have checked that they are breathing normally,



Point their head downward to allow any vomit to escape without them swallowing it or breathing it in.



Lay them on one side (with a cushion at their back if possible),

Bring their knee forward





NOTES

STUDY UNIT 12

SOFT TISSUE INJURIES, INFECTION AND BURNS

LEARNING OUTCOMES

On completion of this study unit the student will....



- ♦ Demonstrate knowledge of the various wounds a first aider needs to treat during a emergency.
- ♦ Demonstrate the necessary skill to treat soft tissue wounds during the vent of a emergency.
- ♦ Define infections.
- ♦ Demonstrate knowledge of the classification of burns.
- ♦ Demonstrate the necessary skill to treat burns.
- ♦ Demonstrate knowledge to manage burns.

1. Types of wounds

- ♦ Bruises
- ♦ Incisions, cuts and lacerations
- ♦ Stab wounds
- ♦ Gunshot wounds
- ♦ Abrasions (Grazes)
- ♦ Avulsions (loose flaps, partial amputation)
- ♦ Amputations

1.1 Bruises

- ♦ Bruises are caused by blunt trauma / injury to tissues, resulting in damage to blood vessels beneath the surface.

TREATMENT:

- ♦ TSCABC! (time, safety, communication, airway, breathing and circulation)

1.2 Incisions, cuts and lacerations

- ♦ These wounds are commonly known as 'gashes, tears or cuts' of the skin. The skin surface is split or torn following blunt trauma, and the force causes the full thickness of the skin to be damaged. Lacerations therefore bleed profusely.

TREATMENT:

- ♦ TSCABC! (time, safety, communication, airway, breathing and circulation)
- ♦ Control bleeding
- ♦ Prevent contamination
- ♦ Suspect internal organ damage and bleeding
- ♦ Suspect shock and treat
- ♦ ALS! (Advance life support) if necessary

1.3 Stab wounds

- ♦ These are wounds where the depth of injury is greater than the length. They penetrate more deeply than slash wounds and tend to come into contact with vital organs in the chest and abdomen.

TREATMENT

- ♦ TSCABC! (time, safety, communication, airway, breathing and circulation).
- ♦ Control bleeding.
- ♦ Check for exit wounds and treat them too.
- ♦ Do not remove impaled objects that are stuck in the wound, stabilize them.
- ♦ Prevent contamination.
- ♦ Suspect internal organ damage and bleeding.
- ♦ Suspect shock and treat.
- ♦ ALS! (Advance life support) if necessary.

1.4 Gunshot wounds

- ♦ The bullet has contacted the skin, and the gases released have passed into the subcutaneous tissues, causing the stellate shaped laceration. There is also some grey/black discoloration and a faint abrasion ring.

TREATMENT

- ♦ TSCABC! (time, safety, communication, airway, breathing and circulation)
- ♦ Control bleeding
- ♦ Check for exit wounds and treat them too
- ♦ Prevent contamination
- ♦ Suspect internal organ damage and bleeding
- ♦ Suspect shock and treat
- ♦ ALS! (Advance life support) if necessary

1.5 Abrasions (Grazes)

- ♦ An abrasion is a superficial injury, commonly known as a 'graze' or 'scratch'. This type of wound damages only the epidermis (uppermost skin layer), and should not therefore bleed.

TREATMENT

- ♦ TSCABC! (time, safety, communication, airway, breathing and circulation)
- ♦ Painful
- ♦ Usually minor bleeding (oozing)
- ♦ Suspect internal injuries and bleeding (shock)
- ♦ Wash with water and soap or diluted antiseptic solution (Savlon etc.)
- ♦ Cover with bandage or antiseptic ointment and gauze

2. Infection



An infection is the detrimental colonization of a host organism by foreign species. In infection, the infecting organism seeks to utilize the host's resources in order to multiply at the expense of the host. The infecting organism, or pathogen, interferes with the normal functioning of the host and can lead to chronic wounds, gangrene, loss of an infected limb, and even death. The host's response to infection is inflammation.

3. Burns

3.1 Classification of burns

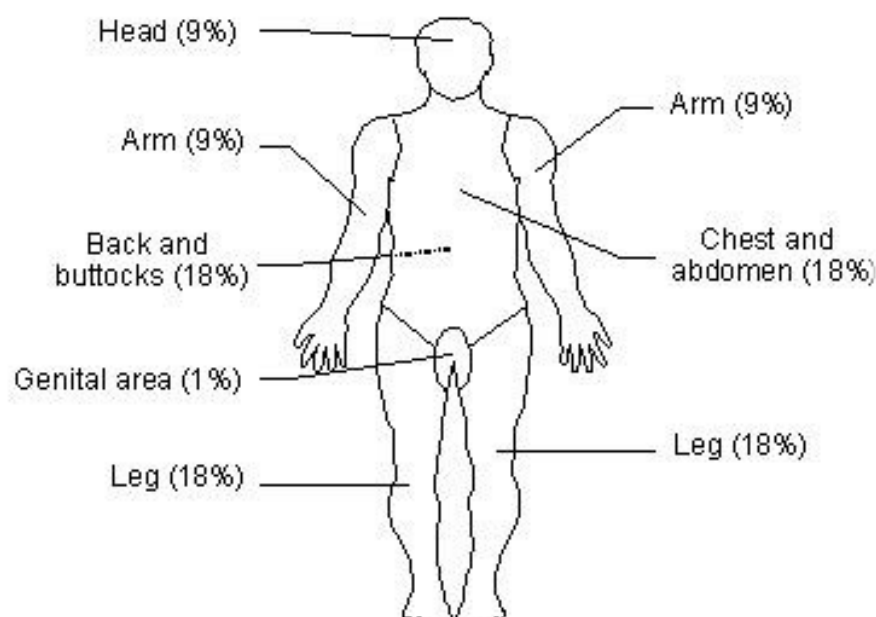
- ♦ Superficial
- ♦ Deep

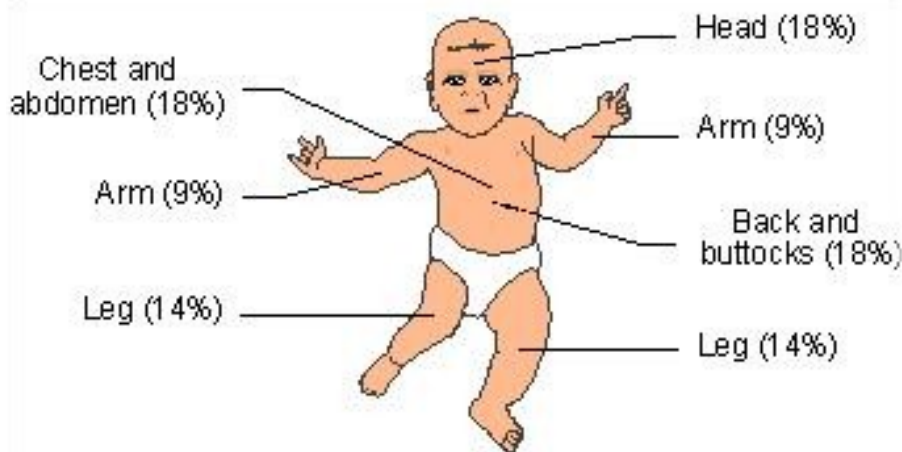
3.2 Causes of burns

- ♦ Moist heat
- ♦ Dry heat
- ♦ Corrosive chemicals
- ♦ Electricity
- ♦ Sunburn

3.3 Surface percentages (rule of nines)

ADULTS



INFANTS**4. Management of burns**

- ♦ TSCABC! (time, safety, communication, airway, breathing and circulation) (watch out for breathing problems when the burns have been caused by fire)
- ♦ Cool with cold running water.(at least 10 minutes)
- ♦ Do not remove anything that sticks to the skin
- ♦ Remove tight clothing and jewellery before swelling starts
- ♦ Always check distal pulse, capillary refill and sensation of burnt limbs
- ♦ Cover burns with sterile dressings (Burnshield)
- ♦ Do not puncture blisters
- ♦ No fluids to drink (wet lips if thirsty)
- ♦ Always suspect shock and treat
- ♦ No ointments or any fatty substances
- ♦ ALS! (Advance life support) if necessary

5. Complications

- ♦ severe infection
- ♦ shock
- ♦ extensive scars
- ♦ loss of function



**The correct treatment is
extremely important!**

6. When to get advance care

- ♦ All burns of more than 3% surface area
- ♦ All deep burns
- ♦ All infants, children and elderly patients
- ♦ Burns to face, hands and genitals
- ♦ Burns complicated by other injuries, bleeding, fractures and contamination
- ♦ All chemical, electrical and lightning injuries
- ♦ Burns to all ill patients



[illegible]

STUDY UNIT 13

FRACTURES, JOINT AND PELVIC INJURIES

LEARNING OUTCOMES

On completion of this study unit the student will....



- ♦ Define a fracture.
- ♦ Demonstrate knowledge of the signs and symptoms indicating that a fracture has occurred.
- ♦ Demonstrate knowledge of the various types of fractures.
- ♦ Demonstrate skill to treat various fractures.
- ♦ Demonstrate knowledge of the various types of splints that can be used.
- ♦ Demonstrate skill on the application of a splint.
- ♦ Define a joint.
- ♦ Demonstrate knowledge of the signs and symptoms of various joints.
- ♦ Demonstrate skill to treat joints in the event of an injury.
- ♦ Define a pelvic injury.
- ♦ Demonstrate knowledge of the signs and symptoms of pelvic injuries.
- ♦ Demonstrate skill to treat a pelvic injury.

1. Fractures

DEFINITION

It is a broken or cracked bone in the body.
.... according to the Oxford English Dictionary it is defined as "the act of being broken."

2. Signs and Symptoms

- ♦ Local pain
- ♦ Local swelling (not always visible)
- ♦ Bruises
- ♦ Limited movement
- ♦ Deformity (not always visible)
- ♦ Shortening (not always visible)
- ♦ Abnormal movement.
- ♦ Irregularity (not always visible)
- ♦ Crepitus (grinding sound of bone ends scraping against each other)

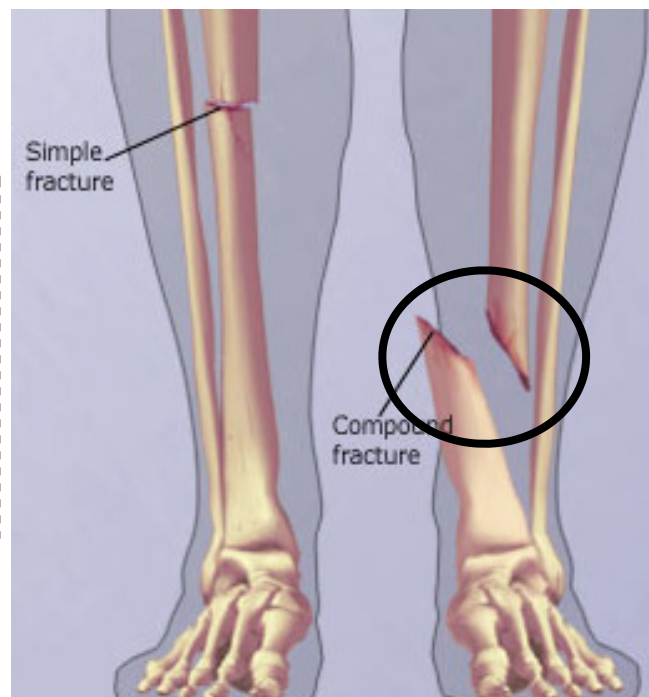
3. Types of Fractures

- ♦ Open Fracture
- ♦ Closed Fracture

3.1 Open Fracture

DEFINITION

An open fracture is a broken bone that penetrates the skin. This is an important distinction because when a broken bone penetrates the skin there is a need for immediate treatment.



Treatment

- ♦ Assess for fracture
- ♦ Immobilize the injury. Any movement will be very painful and can worsen the injury.
- ♦ Clean the wound by thoroughly rinsing it with the disinfected water, using an irrigation syringe.
- ♦ Realign the fractured bone, unless it's a wrist or shoulder fracture.
- ♦ Clean and dress all wounds
- ♦ Remove tight clothing, watches, jewelry: anything that could impede circulation.
- ♦ Check circulation, sensation and motion beyond the fracture site.
- ♦ Splint the fracture
- ♦ Re-check circulation, sensation and motion beyond the fracture site.
- ♦ Elevate the fracture 6-10" to reduce swelling.
- ♦ Apply cold packs or ice to the fracture as long as there is no risk of hypothermia or frostbite.
- ♦ Monitor and treat for shock.



CRITICAL

1. **Don't apply cold directly to the skin: wrap cold packs in socks or other clothing.**
2. **Some fractures, particularly of the pelvis or femur, are frequently associated with severe bleeding. Treat for bleeding as necessary and keep monitoring for shock.**

3.2 Closed Fracture



DEFINITION

A closed fracture is a broken bone that does not penetrate the skin. This is an important distinction because when a broken bone penetrates the skin there is a need for more immediate treatment.

Treatment

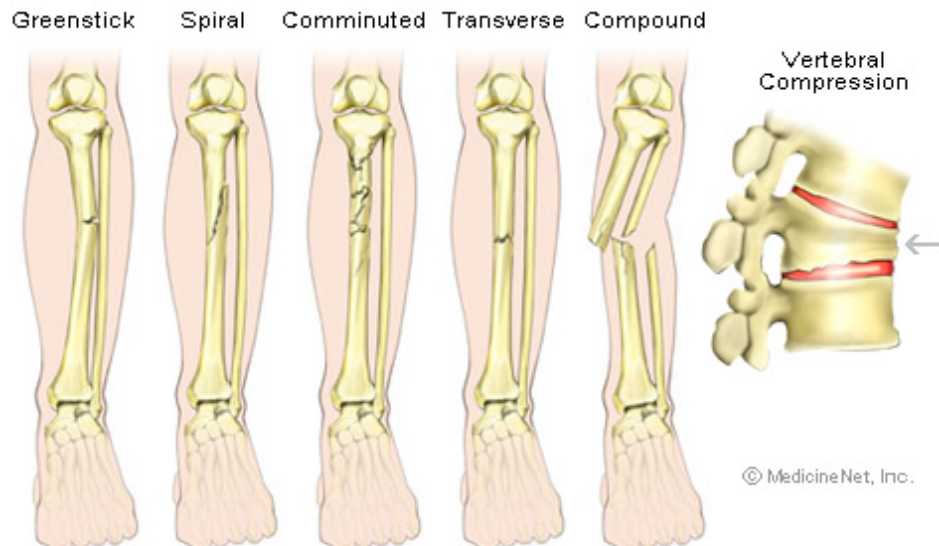
- ♦ Assess for fracture.
- ♦ Immobilize the injury. Any movement is not only very painful but can worsen the injury.
- ♦ Decide whether or not to realign the fractured bone, and realign if necessary.
- ♦ Clean and dress all wounds.
- ♦ Remove tight clothing, watches, jewelry or anything that could impede circulation.
- ♦ Check circulation, sensation and motion beyond the fracture site.
- ♦ Splint the fracture.
- ♦ Recheck circulation, sensation and motion beyond the fracture site.
- ♦ Elevate the fracture 6 to 10 inches to reduce swelling.
- ♦ Apply cold packs or ice to the fracture, as long as there is no risk of hypothermia or frostbite.
- ♦ Monitor and treat for shock.

IMPORTANT

1. Don't apply cold packs directly to the skin: Wrap cold packs in socks or other clothing.
2. Leave the fracture site uncovered: It's important to be able to monitor the area during evacuation.
3. Some fractures - particularly of the pelvis or femur - are frequently associated with severe bleeding. Treat for bleeding as necessary and continue monitoring for shock.

4. Specific fractures

- ♦ Collar bone
- ♦ Shoulder blade
- ♦ Humerus
- ♦ Elbow
- ♦ Radius and ulna
- ♦ Wrist
- ♦ Hand
- ♦ Fingers
- ♦ Femur
- ♦ Knee
- ♦ Tibia and fibula
- ♦ Ankle



Typical Bone Fractures

5. Splints

5.1 Types of splints

- ♦ Hard splints
- ♦ Soft splints
- ♦ Bandages
- ♦ Improvised splints



5.2 Splinting a fracture

- ♦ Assess for fracture.
- ♦ Immobilize the injured leg. Any movement will be very painful and can worsen the injury.
- ♦ Get some rigid material long enough to extend at least from above the knee to below the heel. One possibility is to cut an ensolite sleeping pad to the right length and then fold it over to increase rigidity.
- ♦ Gather materials to tie the splint into place. Any of the following will do: pack straps, rope, clothing, belts, bandannas, triangular bandages.
- ♦ Make sure the leg is straight.
- ♦ Place the splinting material along the underside of the leg, so that it extends from above the underside of the knee to below the heel.
- ♦ Fold the splinting material up around the sides of the leg.
- ♦ Tie the splinting material into place below and above the fracture. Make sure both the knee and ankle are immobilized.
- ♦ Place padding in any empty spaces between the splint and the leg.
- ♦ Check circulation, sensation and motion beyond the fracture site, to make sure the splint isn't too tight.
- ♦ Continue with further treatment for the fracture.



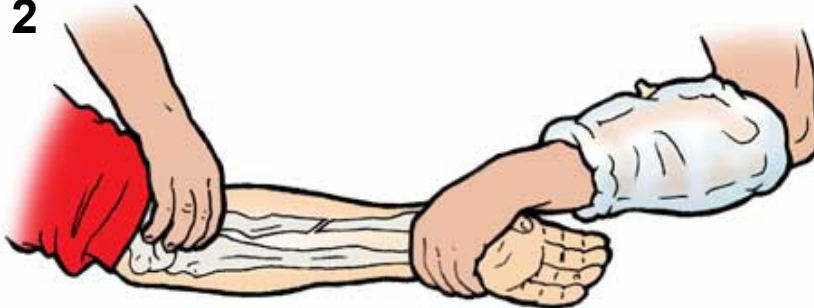


USING A AIR SPLINT

1



2



3



4



TIP

You can also use tent poles, trekking poles, ice axes or pack frames for splinting. Just be sure to put a lot of padding between the splint and the leg.

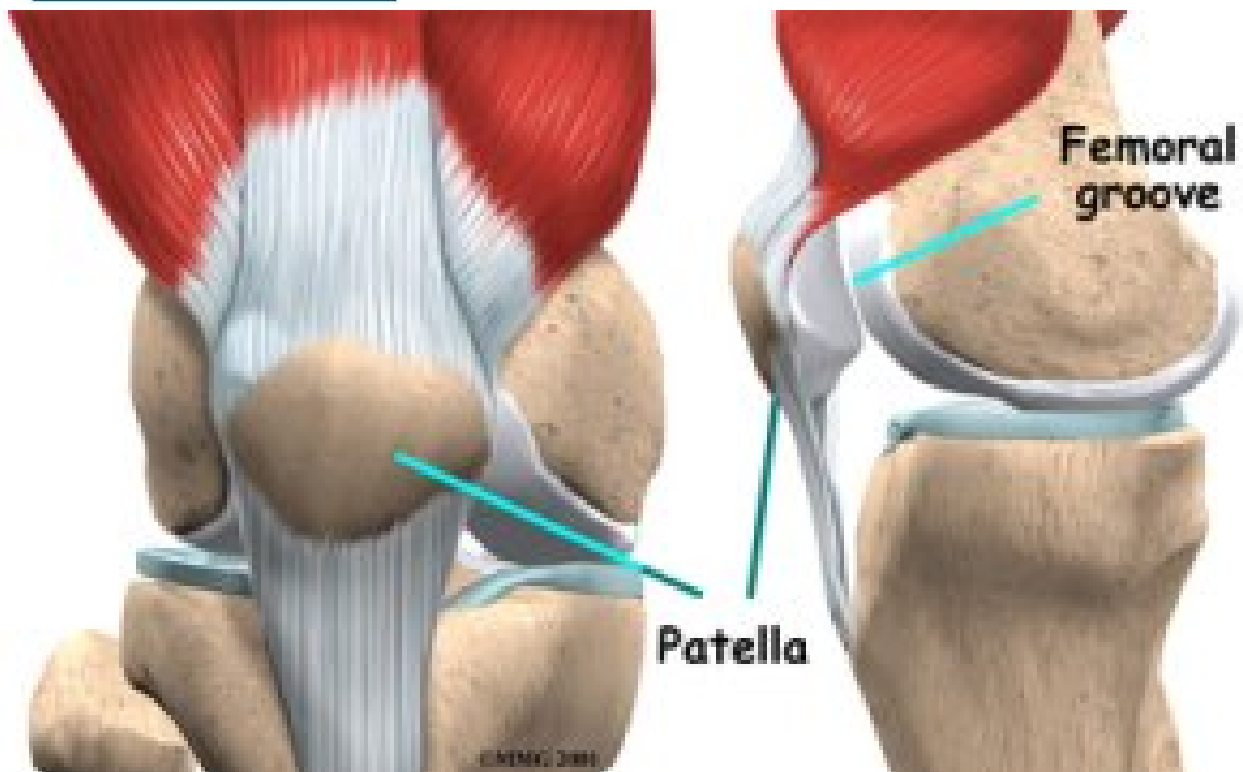
CRITICAL

Don't tie the splint on too tight. This could impede circulation.

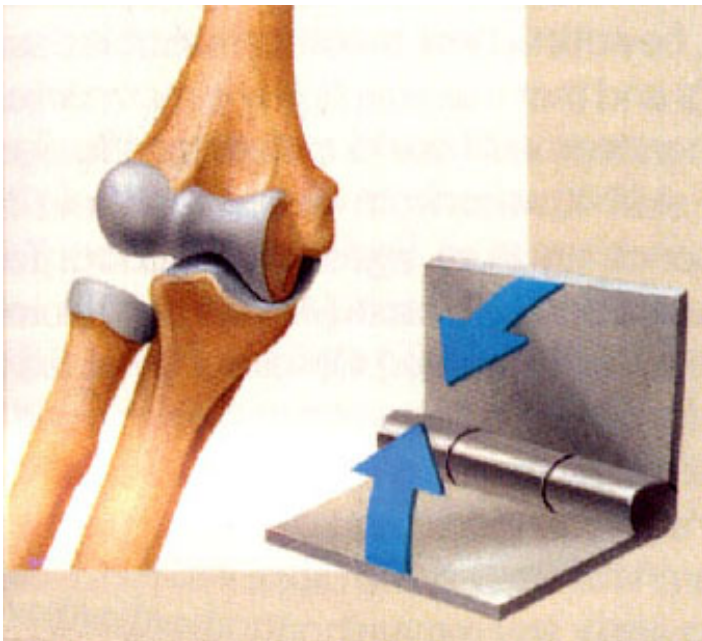
6. Joints

DEFINITION

Joint: A joint is the area where two bones are attached for the purpose of motion of body parts. A joint is usually formed of fibrous connective tissue and cartilage. An articulation or an arthrosis is the same as a joint.

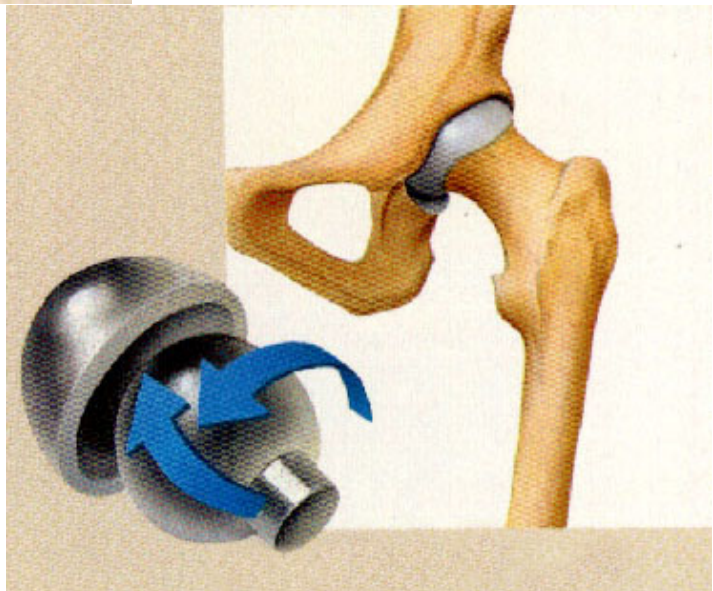


6.1 Types of joints



The hinge joint (as in the knee and elbow)

The ball and socket joint (as in the hip and shoulder)



6.2 Signs and symptoms

- ♦ Local pain.
- ♦ Local swelling.
- ♦ Limited mobility.
- ♦ Unnatural position.
- ♦ Dis-coloration (subcutaneous bleeding)
- ♦ Possible shock.

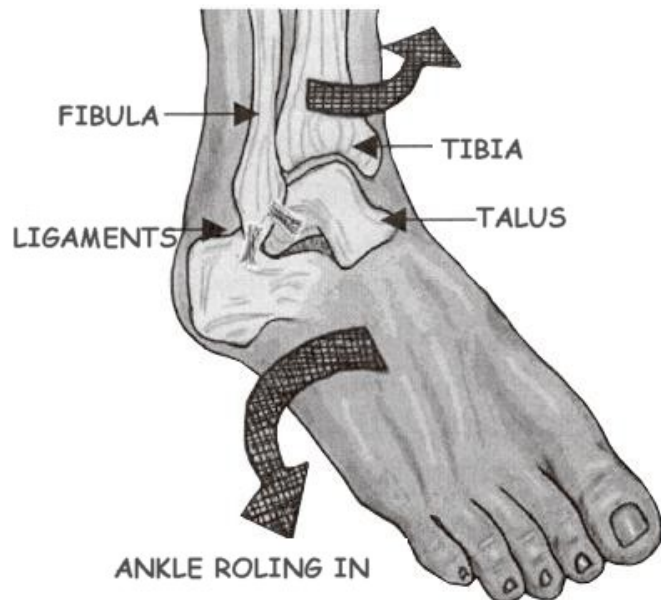
Treatment

- ♦ Splint or immobilise in the position found (or as comfortable as possible) and prevent bending joint.

7. Sprains

DEFINITION

Unnatural movement of a joint, stretching or tearing the ligaments. (May include dislocations)



7.1 Signs and symptoms

- ◆ Local pain.
- ◆ Local swelling.
- ◆ Limited mobility.
- ◆ Dis-coloration.(subcutaneous bleeding)
- ◆ Possible shock.

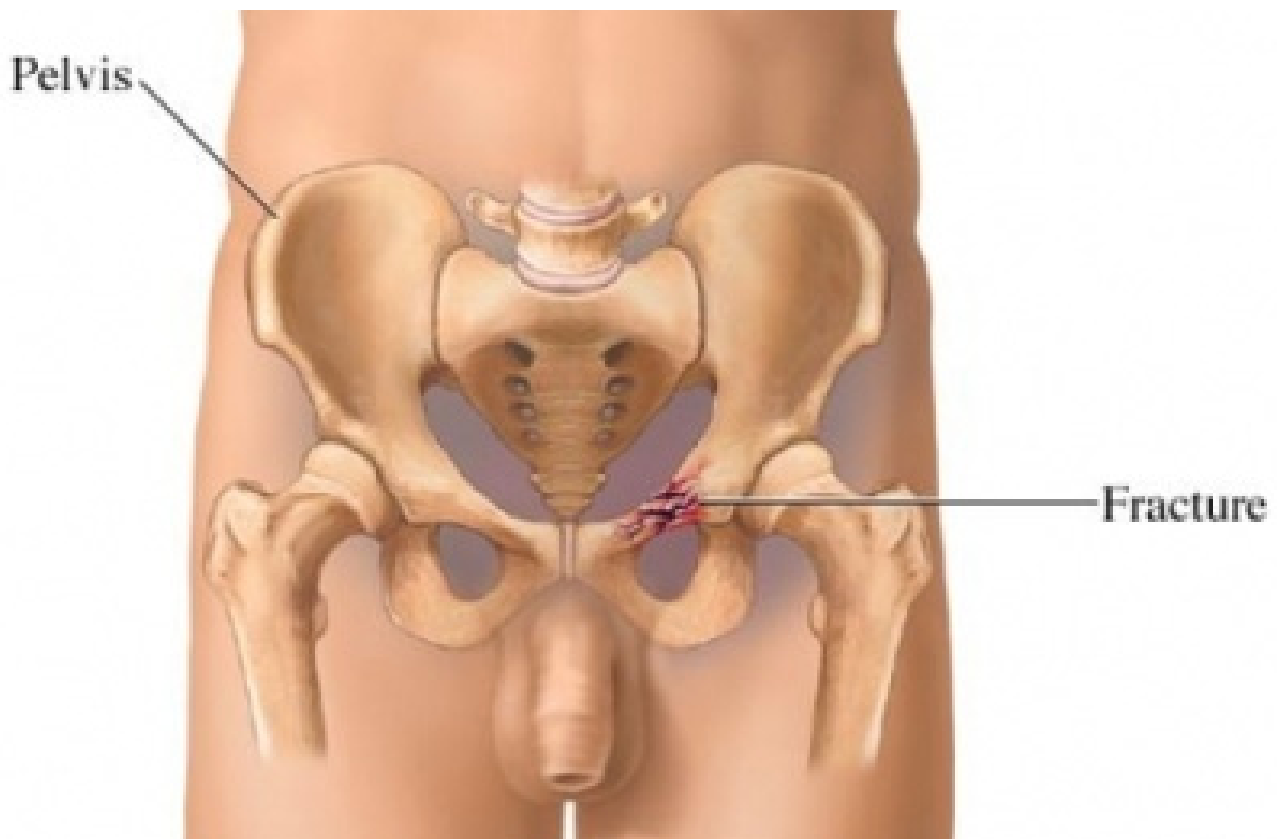
Treatment (RICE)

- ◆ Rest, immobilise and support (splint), try not to move or use joint.
- ◆ Ice, cool sprain in ice water or cold dressing.
- ◆ Compression, apply stretch bandage.
- ◆ Elevate, lift sprain above level of heart.

8. Pelvic injuries

DEFINITION

A pelvic fracture is a break in one or more bones of the pelvis.



8.1 Causes

- ♦ Direct force.
- ♦ Indirect force.(transferred force)

8.2 Signs and symptoms

- ♦ Pain in the pelvic region.
- ♦ Pain with movement of legs.
- ♦ Pain with downward and sideways pressure on pelvic crests and pubis.
- ♦ Foot turned outward when pelvis is fractured.
- ♦ Inability to urinate.
- ♦ Blood in urine.

Treatment

- ♦ TSCABC!
- ♦ Avoid excessive movement at all cost.
- ♦ Position as comfortable as possible.
- ♦ Test distal pulse, capillary refill, mobility and sensation on both legs and compare these on legs.
- ♦ Tie legs and feet together at intervals with padding between legs, if pain increases, don't do it!
- ♦ Never tie around pelvis.
- ♦ Beware of spinal injuries.
- ♦ Use scoop stretcher if available or log roll onto spine board.
- ♦ Suspect shock and treat.

[illegible]

STUDY UNIT 14

BANDAGES AND WOUND DRESSINGS

LEARNING OUTCOMES

On completion of this study unit the student will....



- ♦ Identify various bandages and the use of each.
- ♦ Demonstrate skill in the use of identified bandages on specific injuries.

1. Definition



A bandage is a piece of material used either to support a medical device such as a dressing or splint, or on its own to provide support to the body.

Bandages are available in a wide range of types, from generic cloth strips to specialized shaped bandages designed for a specific limb or part of the body, although bandages can often be improvised as the situation demands, using clothing, blankets or other material.

In common speech, the word "**bandage**" is often used to mean a dressing, which is used directly on a wound, whereas a bandage is technically only used to support a dressing, and not directly on a wound.

2. Purposes of bandages

- ◆ Support and immobilise limbs.
- ◆ Keep splints in position.
- ◆ Keep wound dressings in position.
- ◆ Cover and protect wounds.
- ◆ Control bleeding and swelling.

3. How do I apply a bandage?

The key points when applying a bandage are:

- ◆ Make sure the person is comfortable and tell them what you are doing.
- ◆ Make sure you work from the side of the injury and do not have to lean across their body.
- ◆ Keep the injured part of the body supported in the position it will be in when the bandage is on.
- ◆ Make sure you use the right size bandage as different parts of the body need different widths of bandage.
- ◆ After you have put the bandage on, secure the end by folding it over and tying a knot in the end. You can also use a safety pin, adhesive (sticky) tape, or a bandage clip.
- ◆ As soon as you have put the bandage on, ask the person if the bandage feels too tight and check the circulation by pressing a nail or a piece of skin until it turns pale. If it the colour does not return straight away the bandage may be too tight, so you should loosen it. Limbs can swell up after an injury so check the circulation every ten minutes after you have put the bandage on.

3.1 Main types of bandages:

- ♦ Roller
- ♦ Tubular
- ♦ Triangular

Roller bandages are used to secure dressings and support injured limbs. They can be made of cotton, gauze, elasticated fabric or linen.

There are three types of roller bandage:

- ♦ bandages that are made of open-weave material allow ventilation but do not put pressure on wounds and do not support joints,
- ♦ elasticated bandages mould to the body shape and are used to secure dressings and support soft tissue injuries like sprains, and
- ♦ crepe bandages are used to give firm support to injured joints.

To apply a roller bandage:

- ♦ Keep the rolled part of the bandage above the injury and the unrolled part below the injury.
- ♦ Make two straight turns in the bandage to hold the end in place.
- ♦ Work up the limb, winding the bandage from the inside to the outside in spiralling turns making sure that each new turn covers one third to two thirds of the previous one.
- ♦ Finish with one straight turn and secure the end of the bandage.







**IMPORTANT**

When applying bandages to elbows and knees (to hold dressings in place or support sprains or strains) flex the joint slightly, apply the bandage in a figure of eight and extend the bandage quite far on each side of the joint.

IMPORTANT

When applying bandages on hands (to hold dressings in place or to support sprains and strains), work from the inside of the wrist using diagonal turns across the back of the hand to the end of the little finger, leaving the thumb free.

Tubular bandages are used to hold dressings on fingers or toes or support injured joints. They are made of seamless tubular fabric. You can get elasticated ones for joints such as the ankle. Ones made of tubular gauze can be used for fingers or toes but they do not provide any pressure to stop bleeding.

4. Triangular bandages

DEFINITION

A tube bandage is applied using an applicator, and is woven in a continuous circle.

It is used to hold dressings or splints onto limbs, or to provide support to sprains and strains and to stop bleeding.



Triangular bandages are normal bandages that are folded into a triangle shape. They can be used as large dressings, as slings to support a limb or to secure a dressing in place. You can also use a household item like a scarf to make a triangular bandage. If you are using the bandage as a sling on an arm, you use it opened out.

- ♦ Ask the person to hold their arm across their chest and support the arm while you work.
- ♦ Put the bandage under the arm and around the back of the neck.
- ♦ Put the other half of the bandage over the arm to meet at the shoulder and tie into a knot.

TAKE NOTE

If you are using a triangular bandage to support a limb or large dressing you fold it in half horizontally so the point of the triangle touches the middle of the long edge. Then fold it in half again in the same direction to make a broad strip.

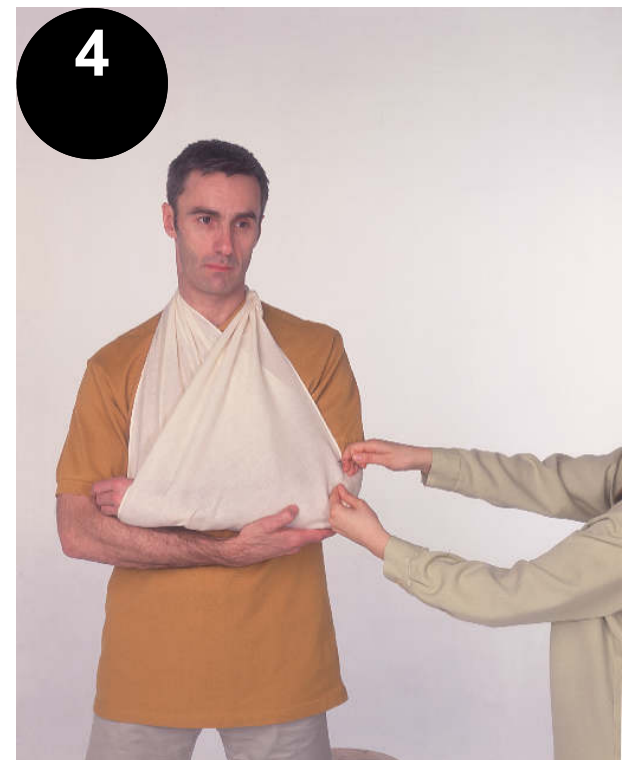
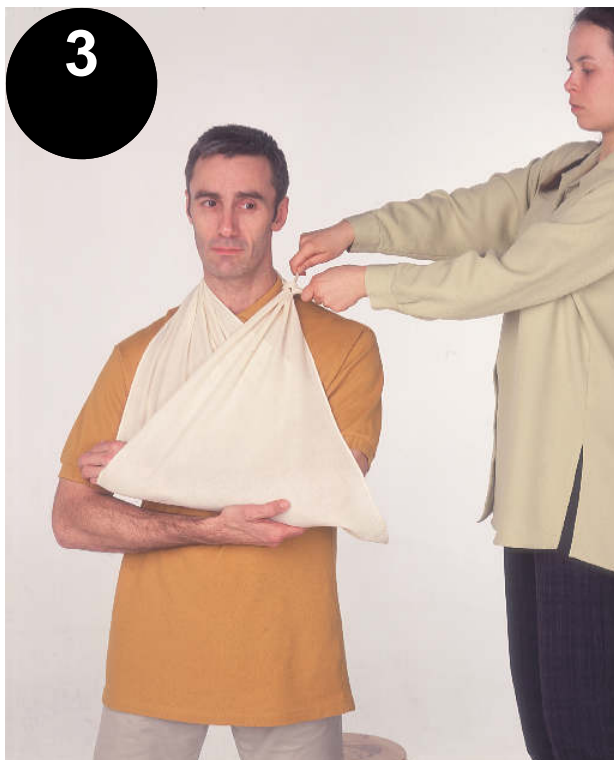
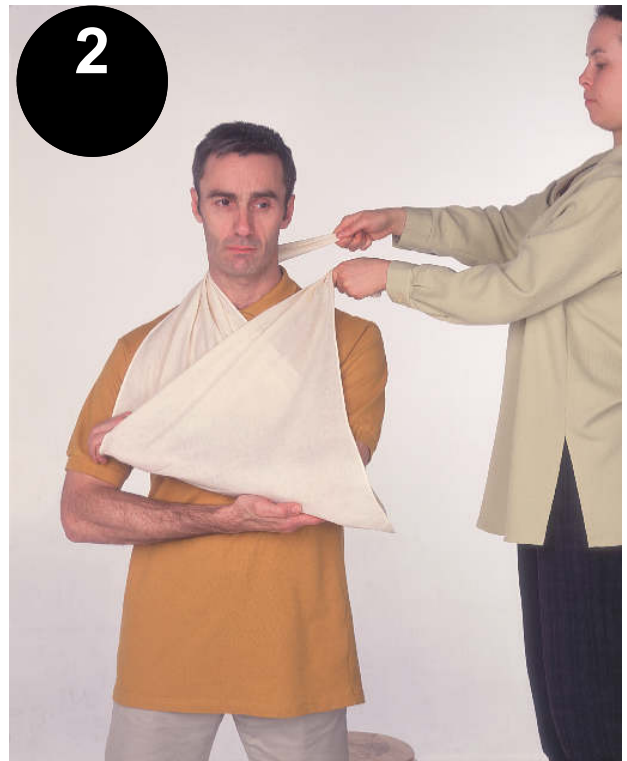
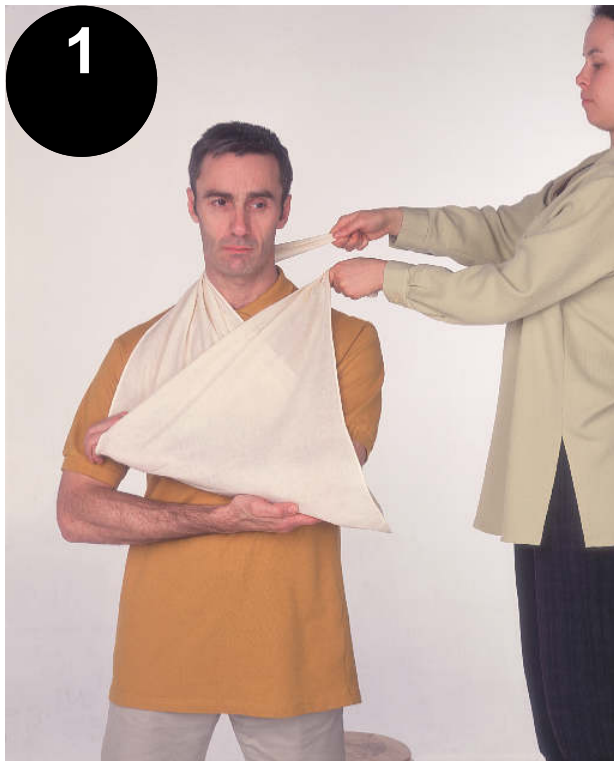


**Applying a triangle bandage:**

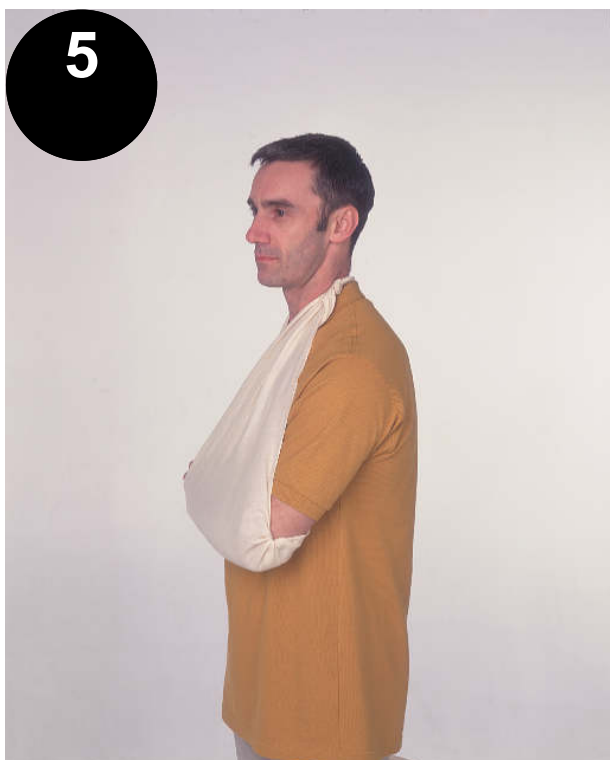
Sometimes an arm is 'put up' with the fingers touching the opposite shoulder. In this case the elbow requires support. The same bandage is used, but the corner is placed on the opposite side.

- ♦ beyond the hand and the lower border or folded side of the triangle is at the elbow ;
- ♦ one end is placed, as before, over the shoulder of the injured side, the bandage is brought over the arm, and the two ends tied at the side of the neck ;
- ♦ the corner is folded up and over the forearm, and pinned to the bandage at the neck on the injured side.





5



5. Gauze bandage

DEFINITION

The most common type of bandage is the gauze bandage, a simple woven strip of material which can come in any number of widths and lengths. A gauze bandage can be used for almost any bandage application, including holding a dressing in place.



Applying a gauze bandage





6. Compression bandage



The term 'compression bandage' describes a wide variety of bandages with many different applications

Short stretch compression bandages are applied to a limb (usually for treatment of lymphedema or venous ulcers). This type of bandage that is capable of shortening around the limb after application and is therefore not exerting ever increasing pressure during inactivity. This dynamic is called resting pressure and is considered safe and comfortable for long-term treatment. Conversely, the stability of the bandage creates a very high resistance to stretch when pressure is applied through internal muscle contraction and joint movement. This force is called working pressure.

Long stretch compression bandages have long stretch properties, meaning their high compressive power can be easily adjusted. However, they also have a very high resting pressure and must be removed at night or if the patient is in a resting position.



The first step to put on the compression device is the placement of the positioner on the lower leg (over the local dressing). Then the device is slipped over the positioner up to the desired position. The positioner is then removed by pulling it down by the handle toward the opening in the foot section. As the positioner is gliding on itself and not on the local dressing, there is no danger that the dressing moves down when pulling off the device.



6.1 General application of compression bandage



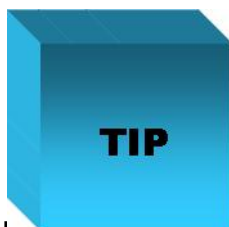
A good bandage should tightly surround the leg on all sides, decrease in pressure from the distal to the proximal side and does not constrict the leg anywhere.

Depending on the size and circumference of the leg, 8 cm or 10 cm wide bandages for dressings are best suited on the lower leg. The bandage is picked up so that the rolled up part of the bandage is on top and points outward. Only in this manner is it possible to unroll the bandage on the leg.

- ◆ Unroll the bandage directly on the skin during application; pull both edges evenly in the direction of application.
- ◆ Never pull the bandage away from the leg because then the guidance is lost and strangling furrows are created.
- ◆ The ankle joint must be kept right-angled to apply the dressing.
- ◆ The compression bandage begins above the metatarso-phalangeal joint. The heel must also be carefully bandaged.
- ◆ The cavities such as the perimalleolar region are padded with hard wadding for pressure equalization.



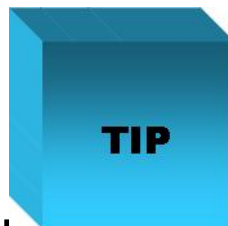
- ◆ At the knee the bandage should include the head of the fibula. Congestion in the hollow of the knee space and thus chafing of the bandage can be avoided if the lower leg is bent at a right angle during bandaging.
- ◆ Generally bandage dressings have a better grip when a second bandage is applied “crosswise” over the first bandage in the opposite direction of application, similar to the well-known Pütter cross bandaging technique.



The lower leg should be bent at a right angle



- ♦ A correctly applied bandage gives the patient the feeling of firm support and is found to be comfortable. Pains should subside. If pains increase or new ones occur which do not disappear while walking around, the bandage must be removed immediately.
- ♦ A compression bandage achieves its full effectiveness only in conjunction with active movement. The more the patient can walk around in the bandage, the greater the success of the treatment will be.



The more the patient can move in the bandage, the greater the success of the treatment will be.



7. Dressings

7.1 How to apply plasters and other dressings?

Before applying any dressing, you should wash and dry your hands. You should also follow the advice outlined below.

- ◆ Make sure that you (or the person who is bleeding) are sitting, or lying down.
- ◆ Tell the person what you are doing as you apply the dressing.
- ◆ If the affected area is bleeding, it should stop if you apply pressure and raise the area higher than the heart.
- ◆ Use a dressing that is slightly bigger than the wound you want it to cover.
- ◆ Hold the dressing at the edges, keeping your fingers away from the part that's going to cover the wound.
- ◆ Place the dressing on top of the wound - don't put it on from the side.
- ◆ A little bit of pressure on the affected area should stop it from bleeding again - but make sure that you don't restrict the circulation.

1



7.2 Sterile dressing pads attached to bandages

Sterile dressing pads come in a protective wrapping. Once they are out of the wrapping, they are no longer sterile (hygienic).

When applying a sterile dressing pad:

- ♦ hold the bandage on either side of the pad,
- ♦ lay the pad directly on the wound,
- ♦ wind the short end once around the limb and the pad,
- ♦ wind the other end around the limb to cover the whole pad, and
- ♦ tie the ends together over the pad to secure it, and put slight pressure on the wound.





7.3 Plasters (also known as adhesive dressings)

Listen to NHS Direct content Plasters are made from a piece of gauze and have an adhesive (sticky) backing. They are usually wrapped in single sterile packs. They sometimes come in different sizes, and shapes, or you can cut them to size. Some plasters are waterproof.

When applying a plaster:

- ♦ clean and dry the skin around the wound,
- ♦ unwrap the plaster and hold it by the protective strips over the backing with the pad side facing down,
- ♦ peel back the strips, but don't remove them, and
- ♦ place the pad on the wound, before pulling away the strips, and pressing the edges of the plaster down.





8. Cotton wool - avoid using on wounds (use for padding with splints)



9. General rules

- ♦ Use clean dressings where possible.
- ♦ Once applied, do not remove, if bleeding continues, bandage over existing dressings.
- ♦ Cover entire wound.
- ♦ Not too loose or tight.
- ♦ Use dry dressings - except in certain circumstances (burns).
- ♦ Tuck away loose ends.
- ♦ Keep fingers and toes open where possible.
- ♦ Test for distal pulse, capillary refill, mobility and sensation on fingers and toes of the injured limbs

CRITICAL

**Always report absence of any of these!
(Possible threatened limb)**

IMPORTANT

Bandages and wound dressings

Test 3 times! (If the arm is involved)

1. Before treating the wound (except in severe bleeding, first stop the bleeding with direct pressure)
2. After covering the wound with a bandage.
3. After putting the arm in a sling (with legs test only twice, as in (i) and (ii) above)

10. Bandaging of specific body parts

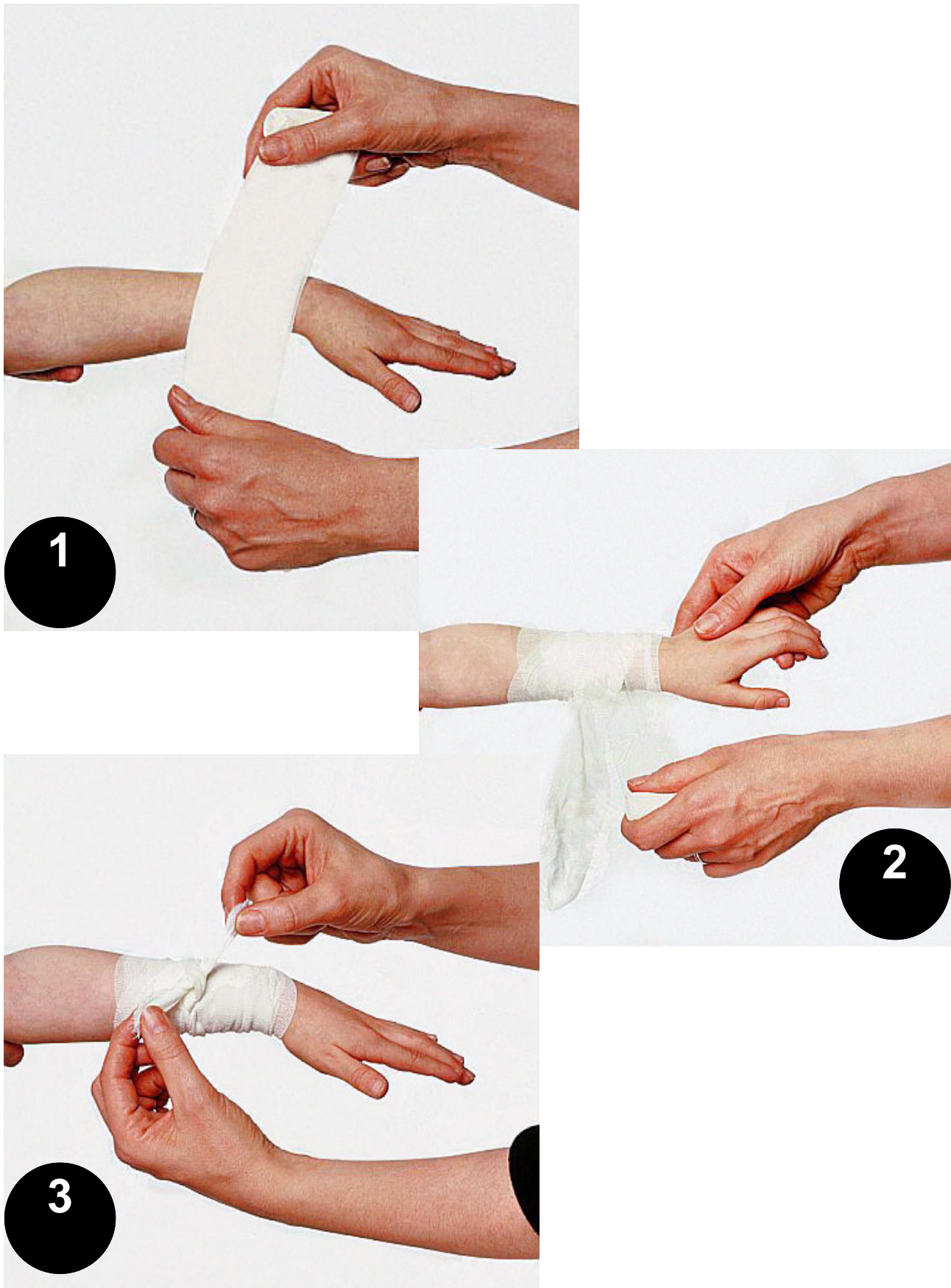
- ♦ Head
- ♦ Eyes and ears
- ♦ Wrist, palm of hand, fingers and fist
- ♦ Elbow
- ♦ Legs (upper and lower)
- ♦ Ankle, feet and toes



10.1 Eyes and ears



10.2 Wrist, palm



10.3 Elbow



10.4 Head



10.5 Legs



10.6 Ankle

This image shows a full page of white paper with horizontal blue or grey ruling lines. The lines are evenly spaced and run across the width of the page, providing a template for writing. There are no margins, text, or other markings present.

STUDY UNIT 15

HEART ATTACKS AND STROKES

LEARNING OUTCOMES

On completion of this study unit the student will....

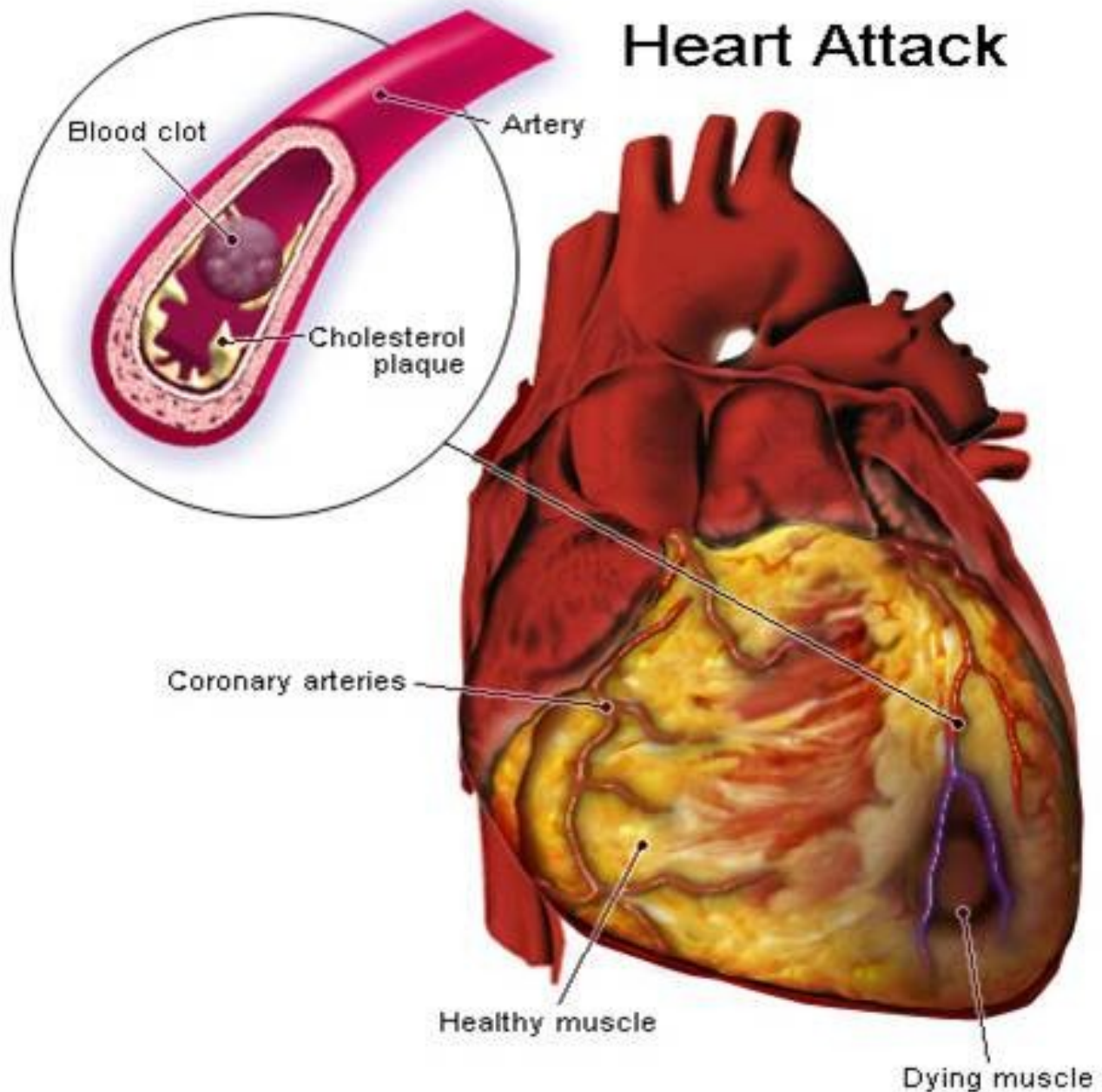


1. Define a heart attack.
2. Demonstrate knowledge of the signs and symptoms of a heart attack.
3. Demonstrate skill in the treatment of a heart attack.
4. Define a stroke.
5. Demonstrate knowledge of the signs and symptoms of a stroke.
6. Demonstrate skill in the treatment of a stroke.

1. Heart attacks

DEFINITION

Complete or partial obstruction of a coronary artery that cause inadequate circulation of oxygen rich blood to the heart muscle which may result in irreversible damage.

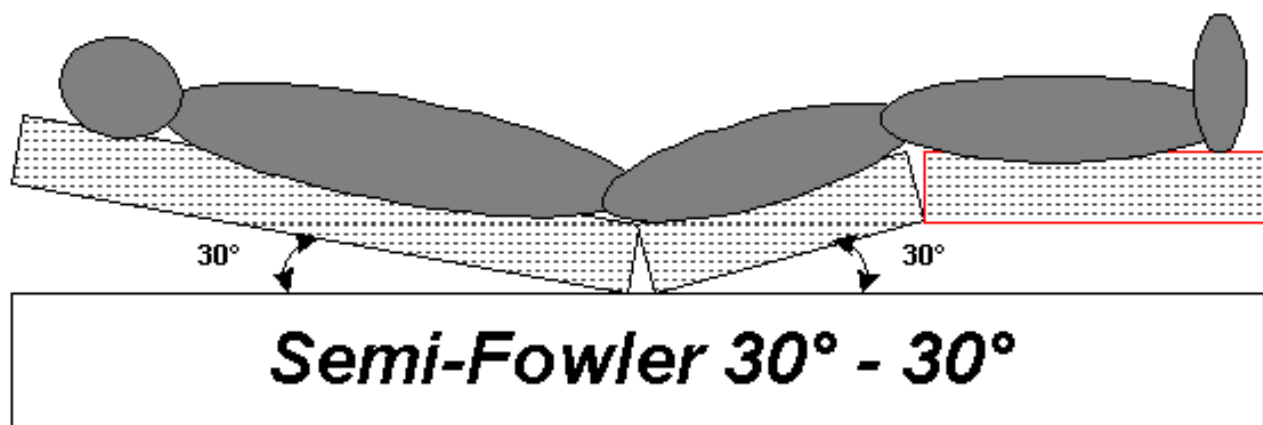


1.1 Signs and Symptoms

- ♦ Severe acute chest pain (may spread to the left jaw, left arm, shoulder blade, pit of stomach, upper back, sometimes feels like indigestion)
- ♦ Sudden weakness and possible fainting
- ♦ Nausea, possible vomiting
- ♦ Excessive perspiration
- ♦ Often laboured breathing and sometimes blue lips
- ♦ Pale, cold skin
- ♦ Rapid, irregular and weak pulse
- ♦ Fear, anxiety
- ♦ Shock (cardiogenic)

Treatment

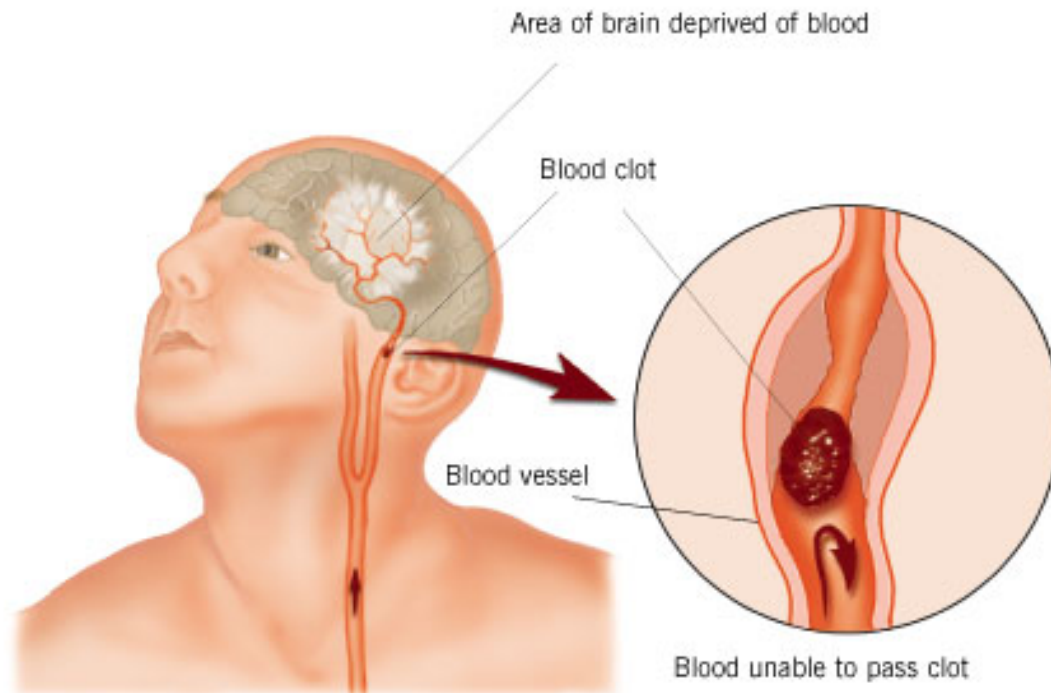
- ♦ TSCABC!! Be prepared for CPR!
- ♦ Always get ambulance, advanced life support (ALS)
- ♦ Semi-fowler position (never on his back or shock position)
- ♦ Loosen tight clothing
- ♦ Reassure patient, keep him still
- ♦ Monitor vital signs regularly



2. Stroke



Complete or partial obstruction of an artery in the brain that causes inadequate circulation of oxygen rich blood to the brain which may result in irreversible damage.



2.1 Signs and symptoms

- ♦ Sudden onset of signs and symptoms.(without warning)
- ♦ Sudden collapse.
- ♦ Lowered level of consciousness.
- ♦ Very confused and disorientated.
- ♦ One-sided weakness or paralysis of limbs.
- ♦ One-sided paralysis of facial muscles.
- ♦ One-sided paralysis of the mouth corner.(sagging)
- ♦ Uncontrolled salivating.
- ♦ Speech difficulty.
- ♦ One-sided visual disturbances.
- ♦ Unequal pupils.
- ♦ Difficult breathing, snoring.
- ♦ Strong, rapid pulse.
- ♦ Loss of bowel and bladder control.
- ♦ Possible convulsions.

Treatment

- ♦ TSCABC!
- ♦ Reassure patient constantly.
- ♦ CPR when necessary.
- ♦ Shock.
- ♦ Semi-Fowler position.
- ♦ Beware of vomiting and aspiration.
- ♦ Recovery position if unconscious.
- ♦ ALS!

[illegible]

STUDY UNIT 16

THE FIRST AID KIT

LEARNING OUTCOMES

On completion of this study unit the student will....



- ♦ Identify the necessary equipment and components that needs to be in a complete first aid kit.

1. Specification of First Aid Kits for 18 Plus Events

Due to concerns regarding Health and Safety Issues at National and Area events these guidance notes have been put together about the recommended contents of First Aid Kits for use by Members taking part in outdoor events.

TAKE NOTE

Please note that these are only guidance notes for 'general purpose' use, where specialist outdoor activities are undertaken the contents may need to be amended in line with recommendations of the appropriate governing body.

General First Aid Kits should be kept in a container made of suitable material and so designed to protect the contents. All boxes should be clearly marked, the recommended marking being a white cross on a green background (Health and Safety (Safety Signs and Signals) regulations 1996). There are two basic types of First Aid Kit.

Standard First Aid Kit: For use at places where a Local Group may meet regularly.

IMPORTANT

NB: Organisers SHOULD already provide this type of kit at National and Area scale events.



Travelling First Aid Kit: For general outdoor use or when travelling. First aid boxes and travelling first aid kits should contain sufficient quantities of suitable first aid materials and nothing else.



2. Standard First Aid Kit

Minimum quantities for low risk establishments and activities may be considered as a general guidance leaflet on first aid.

- ♦ 20 individually wrapped sterile adhesive dressings (assorted sizes) appropriate for the activity (detectable dressings (coloured blue) should be available if catering is to be undertaken)
- ♦ 2 sterile eye pads
- ♦ 4 individually wrapped triangular bandages (preferably sterile)
- ♦ 6 safety pins* (see note)
- ♦ 6 medium sized individually wrapped sterile unmediated wound dressings (approx. 12cm × 12cm)
- ♦ 2 large sterile individually wrapped unmediated wound dressings (approx. 18cm × 18cm)
- ♦ 1 pair of disposable gloves.
- ♦ In situations where mains tap water is not readily available for eye irrigation, sterile normal saline solution (0.9%) in sealed disposable containers should be provided. Once opened they must not be re-used. The use of eye baths/cups or re-fillable containers is not recommended.

2.1 Expanded First Aid Kit

For larger events...

- ♦ First aid manual,
- ♦ Sterile adhesive bandages in assorted sizes,
- ♦ Assorted sizes of safety pins,
- ♦ Cleansing agent/soap,
- ♦ Latex gloves (2 pairs),
- ♦ Sunscreen,
- ♦ 2-inch sterile gauze pads (4-6),
- ♦ 4-inch sterile gauze pads (4-6),
- ♦ Triangular bandages (3),
- ♦ Nonprescription drugs,
- ♦ 2-inch sterile roller bandages (3 rolls),
- ♦ 3-inch sterile roller bandages (3 rolls),
- ♦ Scissors,
- ♦ Tweezers,
- ♦ Needle,
- ♦ Moistened towelettes,
- ♦ Antiseptic,
- ♦ Thermometer,
- ♦ Tongue depressor blades (2),
- ♦ Tube of petroleum jelly or other lubricant.

3. Travelling First Aid Kit

The emphasis is for the contents to reflect the circumstances in which they may be used, but the following at least should be included:

- ♦ general guidance leaflet on first aid
- ♦ 6 individually wrapped sterile adhesive dressings
- ♦ 1 large sterile unmediated dressing (approximately 18cm × 18cm)
- ♦ 2 triangular bandages
- ♦ 2 safety pins
- ♦ individually wrapped moist cleansing wipes
- ♦ 1 pair of disposable gloves.

Eye bath



Scissors

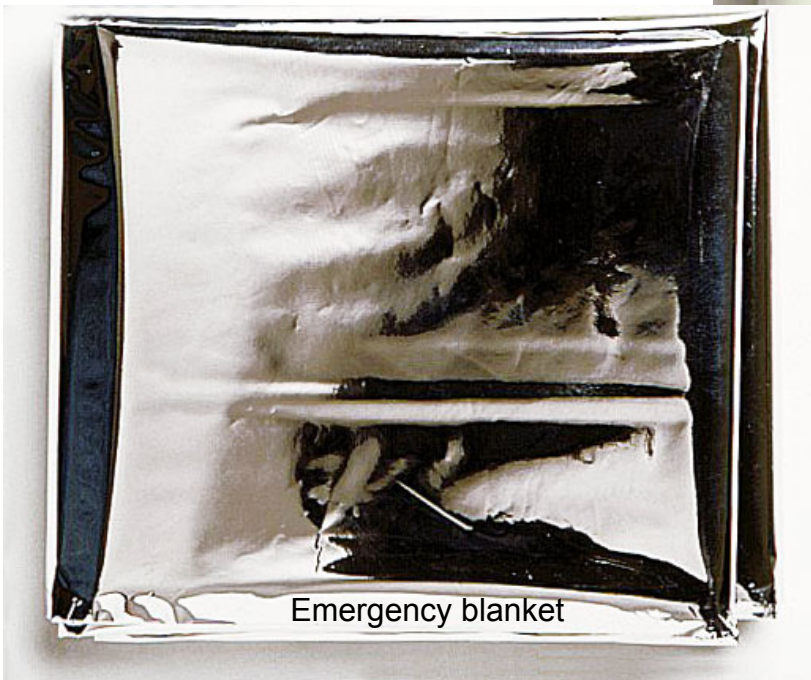


Adhesive bandage

Plasters



Waterproof dressings



Emergency blanket

Sterile dressings and bandage



Antiseptic cream and wipes



Tubular bandage and applicator



Roller bandage





Disposable surgical gloves



Scissors, safety pins and tweezers



Burn dressings



Triangular bandage



Role of cotton and eye pads



Blanket for first aid

4. Care and Replenishment

The contents of first aid boxes and travelling first aid kits should be replenished as soon as possible after use to ensure a continued sufficient supply of materials. All first aid boxes, etc. should be checked regularly to ensure the contents are not used after their expiry date.



It is recommended that nominated person(s) in each group be responsible for maintaining the kit. The first aider may be the appropriate person.

[illegible]

- ♦ RJ VAN WYK First aid training Manual *at the University of South Africa*. Unpublished
- ♦ Die Suid Afrikaanse Noodhulp Handleiding St John Ambulans Vierde Druk
- ♦ <http://www.dkimages.com/discover/Home/Health-and-Beauty/First-Aid/Bandaging/index.html>

Carry out basic first aid treatment in the workplace

SAQA US ID	UNIT STANDARD TITLE		
116534	Carry out basic first aid treatment in the workplace		
ORIGINATOR		REGISTERING PROVIDER	
SGB Occupational Health and Safety			
FIELD		SUBFIELD	
Field 09 - Health Sciences and Social Services		Preventive Health	
ABET BAND	UNIT STANDARD TYPE	NQF LEVEL	CREDITS
Undefined	Regular-Fundamental	Level 3	2
REGISTRATION STATUS	REGISTRATION START DATE	REGISTRATION END DATE	SAQA DECISION NUMBER
Reregistered	2007-08-07	2010-08-07	SAQA 0160/05
LAST DATE FOR ENROLMENT		LAST DATE FOR ACHIEVEMENT	
2011-08-07		2014-08-07	

PURPOSE OF THE UNIT STANDARD

This unit standard is for persons who are responsible to apply first aid in a working place. Persons credited with this standard are able to:

- Describe first aid equipment and explain basic application
- Assess the accident scene, report and make safe
- Prioritise the casualties and treat

LEARNING ASSUMED TO BE IN PLACE AND RECOGNITION OF PRIOR LEARNING

Communications at NQF Level 1, or equivalent competence.

UNIT STANDARD RANGE

The following scope and context applies to the whole unit standard at any working place.

Specific range statements are provided in the body of the unit standard where they apply to particular specific outcomes or assessment criteria.

UNIT STANDARD OUTCOME HEADER

N/A

Specific Outcomes and Assessment Criteria:**SPECIFIC OUTCOME 1**

Describe first aid equipment and procedures, and explain their basic application.

ASSESSMENT CRITERIA**ASSESSMENT CRITERION 1**

The description identifies the dressings and bandages and their application to the specific condition of the injuries in accordance to specific requirements.

ASSESSMENT CRITERION RANGE

Dressings must include adhesive, un-medicated sterile, gauze, roller and pressure bandages, triangular bandages and slings.

ASSESSMENT CRITERION 2

Equipment used in the treatment of the casualty or condition is correctly identified, and the application described according to accepted first aid practice and design specifications.

ASSESSMENT CRITERION RANGE

Equipment must include: stretchers, spinal boards, blankets, splints, and thermometers.

ASSESSMENT CRITERION 3

Personal protective devices for use during treatment of casualties are identified and their purpose and application accurately described and explained.

ASSESSMENT CRITERION RANGE

Protective equipment must include: mouthpieces, surgical gloves.

ASSESSMENT CRITERION 4

The methods for treating injuries and illnesses are described according to international emergency care practice. The correct use and application of equipment with reference to the type of injury/illness sustained are explained.

ASSESSMENT CRITERION 5

The importance of applying methods and procedures is explained with reference to the basic functioning of the body.

ASSESSMENT CRITERION RANGE

The systems must include: respiratory, circulatory, and nervous systems.

SPECIFIC OUTCOME 2

Assess the accident scene of the incident, make safe and report.

ASSESSMENT CRITERIA**ASSESSMENT CRITERION 1**

Workplace hazards and risks are identified. Actions are taken to make safe, and are appropriate to the urgency of the situation and the nature of the hazard or risk according to standard first aid and safety practice.

ASSESSMENT CRITERION 2

Specific hazards and risks associated with the particular incident are identified. Preventative measures are consistent with accepted health and safety practice and particular situational requirements.

ASSESSMENT CRITERION 3

The reporting of the accident/incident to the designated person/s is done in such a way that the nature of incident and support requirements is clearly conveyed.

ASSESSMENT CRITERION 4

The importance of assessing, making safe and reporting is explained with reference to the actual or potential worsening of the incident and working place safety.

SPECIFIC OUTCOME 3

Prioritise casualties and treat.

ASSESSMENT CRITERIA**ASSESSMENT CRITERION 1**

The first assessment of the scene and prioritising of casualties is in line with accepted Triage principles.

ASSESSMENT CRITERION RANGE

Priority order in terms of severity must include: airways, breathing, circulation, internal and external injuries and/or medical conditions.

ASSESSMENT CRITERION 2

People willing and able to assist are identified and tasks assigned according to priority and sound first aid principles.

ASSESSMENT CRITERION 3

The examination is thorough, and accurately determines the nature and severity of the casualty in accordance with sound first aid principles.

ASSESSMENT CRITERION 4

The treatment is consistent with the urgency of the situation, and the nature and severity of the casualty, and aseptic principles.

ASSESSMENT CRITERION 5

The casualty's condition is closely monitored and treatment adjusted as required in the light of vital signs and condition.

ASSESSMENT CRITERION 6

The first aid method applied promotes recovery, and prevent worsening of the condition in accordance with sound first aid principles.

ASSESSMENT CRITERION 7

Reports on the actions taken are clear and promote effective further treatment by emergency support and trained professionals.

ASSESSMENT CRITERION RANGE

Reports to include: Proto Teams, paramedics, medical professionals.

ASSESSMENT CRITERION 8

Details of the incident and treatment undertaken are complete and accurately recorded in the prescribed format.

UNIT STANDARD ACCREDITATION AND MODERATION OPTIONS

Anyone assessing a learner against this unit standard must be registered an assessor with the relevant ETQA. An institution offering learning that will enable achievement of this unit standard must be accredited as a provider through the relevant ETQA by SAQA. Moderation of assessment will be overseen by the relevant ETQA according to the moderation guidelines and the agreed ETQA procedures.

UNIT STANDARD ESSENTIAL EMBEDDED KNOWLEDGE

Essential embedded knowledge will be assessed through assessment of the specific outcomes in terms of

the stipulated assessment criteria. Candidates are unlikely to achieve all the specific outcomes, to the standards described in the assessment criteria, without knowledge of the listed embedded knowledge. This means that for the most part, the possession or lack of the knowledge can be directly inferred from the quality of the candidate's performance. Where direct assessment of knowledge is required, assessment criteria have been included in the body of the unit standard.

The following embedded knowledge is addressed in an integrated way in the unit standard:

- Human anatomy and Physiology
- Emergency Technology

UNIT STANDARD DEVELOPMENTAL OUTCOME

N/A

UNIT STANDARD LINKAGES

N/A

Critical Cross-field Outcomes (CCFO):

UNIT STANDARD CCFO WORKING

Work effectively with others as members of a team, group, organisation or community. Note: The ability and willingness of the candidate to accept, interpret and delegate work instructions correctly, when and if required, in an appropriate manner indicates that he/she can work effectively as a team member in the bigger organisational structure.

UNIT STANDARD CCFO ORGANISING

Organise and manage themselves and their activities responsibly and effectively. Note: The competence in applying acquired knowledge and skills will indicate that the candidate can organise and manage activities in his/her working environment.

UNIT STANDARD CCFO COLLECTING

Collect, analyse, organise and critically evaluate information. Note: The ability of the candidate to collect, organise and evaluate the necessary information and explain meanings and results will indicate proficiency.

UNIT STANDARD CCFO COMMUNICATING

Communicate effectively, using visual, mathematical and / or language skills in the modes of oral and / or written presentations. Note: The ability of the candidate to use assessment skills, communication and language skills will indicate his/her effectiveness to communicate information in the modes of oral and written presentations.

UNIT STANDARD CCFO SCIENCE

Use science and technology effectively and critically showing responsibility towards the environment and health of others. Note: The ability of the candidate to use science and technology effectively will contribute towards promotion of holistic emergency care and rehabilitation.

UNIT STANDARD CCFO DEMONSTRATING

Demonstrate an understanding of the world as a set of related systems by recognising that problem-solving contexts do not exist in isolation. Note: The ability of the candidate to identify and refer anomalous conditions to specialist services confirms understanding that a specific observation, inference, action or decision can have an interrelated effect.

UNIT STANDARD ASSESSOR CRITERIA

Assessors should keep the following principles in mind when designing and conducting assessments against this unit standard:

- Focus the assessment activities on gathering evidence in terms of the main outcome expressed in the title to ensure assessment is integrated rather than fragmented. Remember we want to declare the person competent in terms of the title. Where assessment at title level is unmanageable, then focus assessment around each specific outcome, or groups of specific outcomes.
- Make sure evidence is gathered across the entire range, wherever it applies. Assessment activities should

be as close to the real performance as possible, and where simulations or role-plays are used, there should be supporting evidence to show the candidate is able to perform in the real situation.

- Do not focus the assessment activities on each assessment criterion. Rather make sure the assessment activities focus on outcomes and are sufficient to enable evidence to be gathered around all the assessment criteria.
- The assessment criteria provide the specifications against which assessment judgements should be made. In most cases, knowledge can be inferred from the quality of the performances, but in other cases, knowledge and understanding will have to be tested through questioning techniques. Where this is required, there will be assessment criteria to specify the standard required.
- The task of the assessor is to gather sufficient evidence, of the prescribed type and quality, as specified in this unit standard, that the candidate can achieve the outcomes again and again and again. This means assessors will have to judge how many repeat performances are required before they believe the performance is reproducible.
- All assessments should be conducted in line with the following well documented principles of assessment: appropriateness, fairness, manageability, integration into work or learning, validity, direct, authentic, sufficient, systematic, open and consistent.

UNIT STANDARD NOTES

Specified requirements include legal and site-specific requirements and are contained in one or more of the following documents:

Legal Requirements:

- Resuscitation Council protocol of Southern Africa
- Mine Health and Safety Act.
- Occupational Health and Safety Act
- Chief Inspector of Mines` Directives

Site-specific Requirements:

- Hazard Identification and Risk Assessments (HIRA)
- Occupational Health and Safety Risk Management Programme
- Managerial Instructions
- Mine Standard Procedures
- List of Recorded OH&S Risks
- Working Guides
- Equipment and Materials Specifications